Dr. Sunita Mishra
Ph.D., Indian Institute of Technology Delhi
Assistant Professor
Department of Mining Engineering

भारतीय प्रौद्योगिकी संस्थान खड़गपुर
खड़गपुर, पश्चिम बंगाल, भारत
Indian Institute of Technology Kharagpur
Kharagpur, West Bengal, India

Saturday, 19th October 2024

To,
The Director Technical (RD&T),
Central Mine Planning & Design Institute Limited
(A Subsidiary of Coal India Limited),
Gondwana Place, Kanke Road,
Ranchi – 834 031, Jharkhand, India

Subject: Submission of a revised project proposal titled "An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: A Case Study on the Dragline" for Coal SnT Grant.

Ref: CMPDI/S&T/2023-24(69)/169

Dear Sir,

I am highly thankful to you for considering our proposal and giving us your valuable comments for strengthening the objectives of the project.

Kindly find attached herewith the revised proposal titled "An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: A Case Study on the Dragline" by Prof. Aurobinda Routray, Prof. Sunita Mishra, Prof. Mamata Jenamani and Prof. Sunil Manohar Dash submitted for seeking possible funding opportunities from Coal Science and Technology Research Grant. We have tried to address the review comments/ suggestions given by your esteemed panel. The modifications made to the proposal in accordance with your comments are as follows:

Comment 1: Specific applications, usefulness and details of the equipment on which the proposed sensors will be mounted.

Answer: This system is designed for AI-enhanced diagnosis, prognosis, and health monitoring of electrical motors within various mining electrical systems. It will provide predictive insights regarding potential motor failures or malfunctions in components powered by these motors. The objective of this study is to develop a strain and thermal-based digital twin of the motor, incorporating suitable networking solutions for mining environments. Initially, the proposed system will be installed and evaluated on the electrical system of a company-owned dragline, in collaboration with Northern Coalfield Limited, Singrauli.

Comment 2: Consent letter from any coal-producing company for association in the project as a sub-implementing agency

Answer: The consent letter from Northern Coalfield Limited has been obtained who will be helping in development and implementation of the project deliverables.

Room No.: 105, Department Main Building, Department of Mining Engineering, Indian Institute of Technology Kharagpur, Kharagpur, West Bengal, India – 721 302

Email: smishra@mining.iitkgp.ac.in; sunita.mishra256@gmail.com, Phone: +91-3222-214824 (O), +91-3222-280208 (R), +91-9717253877 (M)

Comment 3: Consent from OEM for installation of proposed sensors in the equipment.

Answer: The consent of OEM is not required as the sub-implementing agency has offered to implement the above sensors in their company owned Draglines.

Comment 4: Consent from DGMS for the proposed system needs to be obtained before submission of the revised proposal.

Answer: The proposal has been modified to test and implement the sensors mainly in surface mining machinery starting with Northern Coalfield Limited where they do not have underground mines. Their key interest was to test and implement the system in their company owned Dragline, which has a maximum number of motors and would yield viable results.

Comment 5: The proposed time for completion shall be reduced to 2 years and accordingly, the proposed outlay along with provisioning under the head "salary and allowances" should be revised.

Answer: The proposed time has been reduced to 2 years from 3 years as suggested by the committee.

Comment 6: Permission from DGMS is required for trials of the proposed system in the underground mines.

Answer: Since installation of sensors in underground mines is not included in the present study, hence, the PI and his team did not approach DGMS for their kind consent. The PIs intends to get the DGMS approval in the next phase of the project.

Comment 7: Desktop, laptop, workstation is generally not permitted from the S&T fund of MoC and therefore, needs to be removed in the revised proposal.

Answer: Desktop, laptop and workstations have been removed from the revised proposal as suggested by the esteemed committee.

Comment 8: Association letter from an industry partner is required with a certain percentage of contribution in the project.

Answer: A support letter from an industry partner is obtained and submitted along with the revised proposal.

Comment 9: Intellectual Property Rights (IPR) has to be shared between CIL (funding agency) and the Principal Implementing / Sub-implementing Agency(s) in equal proportion (50:50) after successful completion and before commercialization of the product/ process developed. In case, there are more than one implementing agencies, the proportion, i.e. 50% of the IPR may be distributed among all implementing agencies in appropriate proportion. Acceptance in this regard, needs to be submitted by the project proponent.

Answer: The PIs agree to the IPR sharing terms and conditions as laid by CIL.

Comment 10: Revised proposal shall be formulated as per S&T guidelines of MoC (available at URL https://scienceandtech.cmpdi.co.in).

Answer: The revised proposal has been submitted in accordance with the S&T guidelines of MoC.

The PIs hope that the queries and comments given by the SSRC committee has been addressed appropriately. I would also like to add that I, Dr. Sunita Mishra, will be the Project Coordinator. Kindly revert back to me in case of any further clarifications. We look forward to receiving suggestions and comments from you.

Thank you.
Sincerely,



Co-PI: Sunita Mishra, Ph.D.
Assistant Professor
Department of Mining Engineering
Indian Institute of Technology (IIT) Kharagpur
West Bengal State, India.
Email: smishra@mining.iitkgp.ac.in
Ph: +91-9717253877

PI: Aurobinda Routray, Ph.D.
HAG Professor
Department of Electrical Engineering
Indian Institute of Technology (IIT) Kharagpur
West Bengal State, India.
Email: aroutray@ee.iitkgp.ac.in
Ph: +91-3222-283072

Co-PI: Mamata Jenamani, Ph.D.
Professor
Department of Industrial and Systems Engineering
Indian Institute of Technology (IIT) Kharagpur
West Bengal State, India.
Email: mj@iem.iitkgp.ac.in
Ph: +91-3222-283740

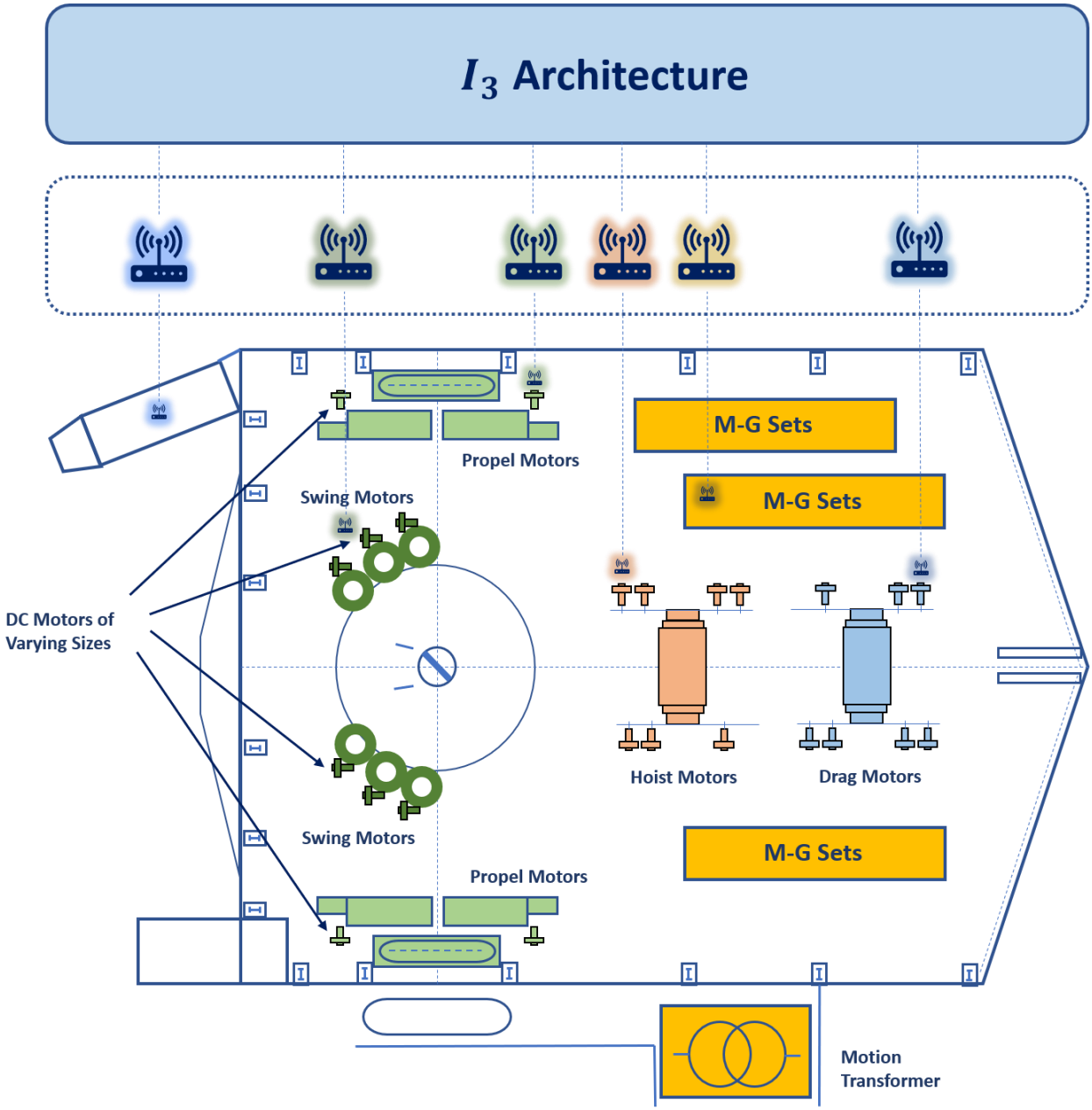
Co-PI: Sunil Manohar Dash, Ph.D.
Assistant Professor
Department of Aerospace Engineering
Indian Institute of Technology (IIT) Kharagpur
West Bengal State, India.
Email: smdash@aero.iitkgp.ac.in
Ph: +91-3222-284512

FORM – I

PROJECT PROPOSAL FOR S&T GRANT OF MoC

1	PROJECT TITLE		An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: A Case Study on the Drag Line
2	Name and address of principal Implementing Agency(s) Name of Project Leader/Coordinator/Principle Investigator		Aurobinda Routray, Professor, Electrical Engineering, IIT Kharagpur
3	Name and address of Sub Implementing Agency(s) Name of Co-investigator(s)		**Implementing Agency**: Northern Coal Fields Ltd, Singrauli **Co-Investigators** 1. Professor Sunita Mishra, Mining Engineering, IIT Kharagpur 2. Professor Mamata Jenamani, Industrial Systems Engineering, IIT Kharagpur 3. Professor Sunil Manohar Dash, Aerospace Engineering, IIT Kharagpur
4	Definition of the issue (Max. 300 words)		This project aims at design, development and field implementation of an AI enabled IoT platform for prognosis, diagnosis and health management of Mining Electrical Equipment. In phase one it has been decided by mutual agreement to implement the system on a typical **drag line** for proving the concept. Northern Coal Field Ltd Singrauli has agreed to the field trial in the first phase. The system will operate at three levels as follows: Level-1: Sensing and Edge computing to detect anomalies and events. It would communicate to Level-2: Would be a local system which, if required can be bypassed for specific applications. This would bring in data for a cluster of sensors from various subsystems. It would create a local interface and database. Level-3: Would operate on cloud. It will make generic decisions based on longitudinal and cross-sectional data across similar subsystems. It would use a trainable degradation model to predict the wear and tear, RUL and do the maintenance scheduling, while generating regular health cards for critical subsystems. The focus of our research would be to create an automatic and systematic plan for monitoring, diagnosis, prognosis and degradation model for the electric motor drives and their controllers. We plan to execute the project in multiple and parallel threads such as: 1. Gathering field data and priors from the Domain Knowledge Experts (DKE) 2. Physics based system model development to understand various fault signatures due to vibration, temperature, current, voltage, power to name a few 3. Machine Learning and Signal Processing for the edge devices to detect events and diagnose the

			faults 4. Design and testing of Sensor nodes 5. Development and testing of Cloud based algorithms for degradation and RUL prediction 6. Field Deployment 7. Final Product Design for TRL-8



Architecture for prognostics and diagnostics of electric motor and drives

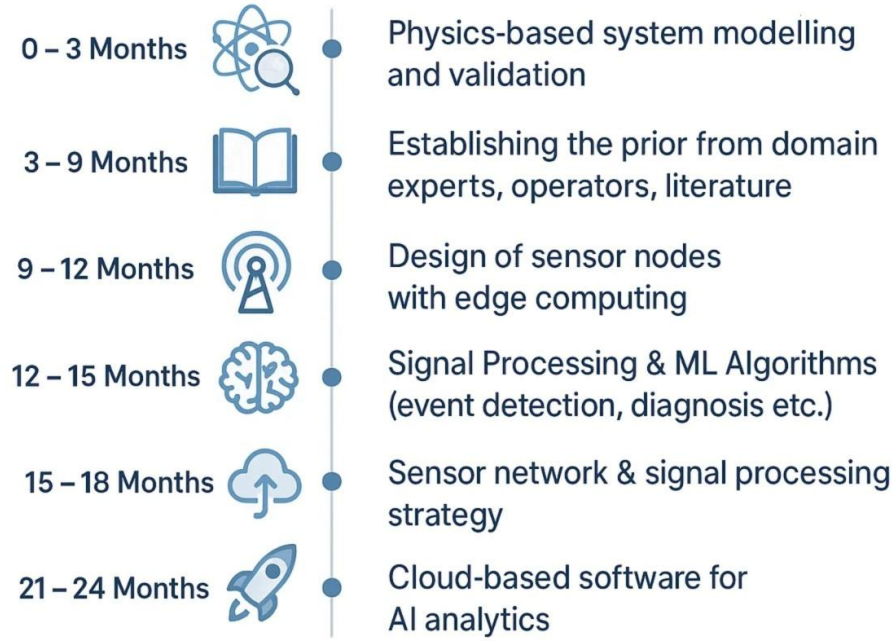
Hence, we propose implementing a proactive maintenance approach, such as utilizing AI and ML-based fault diagnostics and prognostics systems, which can help identify faults, predict failures and recommend proactive maintenance strategies.

5	Objectives (Specific and not more than 2-3)	: -	**Modelling**: Conducting a comprehensive study of the system and creating a detailed model for the system on the basis of electromagnetics, thermal, strain-field and statistical models to generate data. **Hardware Design**: Design and implementation of the sensors and the communication network and deploying a signal processing strategy to gather and analyze data from the system. **Software Design**: Developing AI and machine learning algorithms for event detection, diagnosis, prognosis and RUL prediction of faults
6	Justification for subject area (Max. 200 words)	:	In situ diagnostics and prognosis has become a defacto standard for process industries. For mines it is just the beginning of an era of Industry 4.0 where all static/moving equipment will be connected and monitored by stake holders in a hierarchical fashion. This would minimize the downtime and enhance utility in a cascaded manner. The major production bottle necks can be easily identified by these networked edge-devices. The streaming data and events from these devices can be used to optimize the operation and maximize the production. The equipment in the coal mines operate under harsh ambient conditions, such as high temperature, extreme air quality and long work schedules. Under these conditions the electrical subsystems such as the motor and the associated electronics controllers undergo thermal as well as mechanical stress. They need more frequent periodic maintenance. The proposed system will continuously monitor the conditions of these vulnerable yet important subsystems by way of a low-cost sensing and communication architecture. This will be like a watchdog to identify incipient and weak faults before they turn into major failures. Our system will use extensive signal processing and AI to do data driven as well as model-based diagnostics of these systems in real-time.
7	How the project is beneficial to coal industry		• There will be multi-fold utility and benefits of putting these sensor nodes with edge computing. The obvious advantages are • In situ real-time health monitoring • Efficient condition-based maintenance scheduling Reduction of equipment down-time • Foreseeing the failures and planning accordingly • The real-time data can be used to corroborate production volumes and employee engagements • Localization of fault and ease of diagnosis due to the availability of time-stamped data • Development of accurate degradation models and estimation of RUL.
8	Work Plan (Max. 100 words)	:	The following work elements mark our planning to execute the project 1. W.E.1: Physics based system modelling and validation (Qtr1,2,3) 2. W.E.2: Establishing the prior from domain knowledge experts, operators, opinionated data and literature survey. (Qtr1, Qtr2) 3. W.E.3 Design of sensor nodes with edge computing (Qtr1 to Qtr6)

			4. W.E.4: Design, Coding and Testing of Signal Processing and Machine learning Algorithms for event detection, anomaly detection, diagnosis and prognosis of faults. (Qtr1-Qtr7) 5. W.E.-5: Development of sensor network and signal processing strategy. (Qtr1-Qtr8) 6. W.E.-6 Cloud based software for AI based analytics. (Qtr3-Qtr9) 7. W.E.-7 Deployment and Testing of the prototype. (Qtr6- Qtr10)
8.1	Methodology (Max. 200 words)	:	**WE-1: Physics based system modelling and validation** - Study of the mechanical, electromechanical and control systems • Electromagnetic Model • Thermal Model • Dynamic Strain Field Model • Statistical and Ensemble models **WE-2: Establishing the prior from opinionated data and literature survey–** • On-site Data collection • Field visit and model identification • Opinionated Data collection and verification from extensive literature survey **WE-3: Design of sensor nodes with edge computing** - Sensor Selection and interface to 32-bit devices - Event detection algorithms and communication - Testing simple signal processing for detecting and classifying the faults • Evaluating the performance of these systems in terms of accuracy and speed **WE-4: Design, Coding and Testing of Signal Processing and Machine learning/Deep Learning Algorithms and validation of fault model.** • Time domain, Frequency domain and Time frequency-based signature analysis for steady state and transient analysis under healthy and faulty conditions Development of Machine Learning for feature-based fault detection and classification • Deep Learning Architecture for fault detection and classification **WE-5: Development of sensor network and communication strategy** • Testing the feasibility of several data communication strategy • Communication with local host and cloud • Communication strategy and data transfer with cell phone apps **WE-6: Cloud based software for AI based analytics** - Design of Health Card • Development of degradation model • Strategy for learning the degradation model with time **WE-7: Deployment and Testing of the prototype.** • Design of prototype • Field and Lab testing

8.2	Organization of work elements (Max. 200 words)	:	**For timeline, please check 8.3** The work will be organized using the following aspects Human Resources PI and Co-PI, RS, TA, Interns, Temporary Workforce Lab Resources (LR) PI: Prof. Aurobinda Routray [AR] Co-PI Prof. Sunita Mishra [SM] Co-PI: Prof. Mamata Jenamani [MJ] Co-PI: Prof. Sunil M Dash [SD] Research Scholar [RS-1 RS-2 RS-3] Technical Assistant [TA-1 TA-2] System Manager (SA) Interns [IN-1 IN-2…] Temporary Workforce [TWF-1 TWF-2…] WE-1: Modelling Team: SM (Lead), SD, AR, RS-1, RS-2, TA-1, IN-1, TWF-1: important tasks to be carried out by this team - Study of System operations - Field visit and meeting DKE - Historical data analysis - Steady State modelling and validation - Transient modelling and validation - Deep Learning based Models - Failure models of important subsystems WE-2 Literature Survey and Sentiment Analysis Team: MJ (lead), AR, SM, SD, RS-1, RS-2, RS-3 - Comprehensive Literature review on Shovel Faults and Diagnosis - Even Detection methods, Fault Classification methods - Designing Knowledge graphs from opinionated data from the experts by way of several rounds of discussions WE-3 Design of sensor nodes with edge computing Team: AR (lead), RS-1, RS-2, TA-1, TA-2 This is the hardware design part. Design of three prototypes of sensor nodes based on different processors, 32-bit ARB, RasPI, and NUC, with communication to cloud WE-4 Design, Coding and Testing of Signal Processing and Machine Learning Algorithms and validation of fault model. Team: MJ (lead), AR, SM, SD Level 1 Algorithms at the edge nodes Level 2 Algorithms at the server Level 3 Algorithms at the cloud Testing with historical field data WE-5: Development of sensor network and communication strategy AR (lead), RS-1, RS-2, TA-1, TA-2 Three strategies will be tested Strategy-1: Using IEEE 802.11 (Common wifi)

			Strategy-2: SMS: Short messages Service Strategy-3: LORA WE-6 Cloud based software for AI based analytics MJ (Lead) RS-3, TA-2, IN-2, IN-3 WE-7 Deployment and Testing of the prototype SM (Lead), AR, MJ, SD, RA-1, RA-2, RA-3, TA-1, TA-2 Lab based testing Field testing
8.3	Time schedule of activities giving Milestones (also include a Bar Chart/PERT Chart)	:	**Attached**



9	**Details of proposed outlay** (Rs.in INR lakhs)					
	Sl. No.	Items		Total cost estimated		
				Total project cost	1st Year	2nd Year
		Capital Expenditure				
	9.1	Land & Building	:	0	0	0
	9.2	Equipment	:	₹2,00,00,000	₹2,00,00,000	0
	9.3	***Total Capital*** *(9.1+9.2)*	:	₹2,00,00,000	₹2,00,00,000	0
		Revenue Expenditure				
	9.4	Salaries / allowances	:	₹52,80,000	₹26,40,000	₹26,40,000
	9.5	Consumables	:	₹14,00,000	₹7,00,000	₹7,00,000
	9.6	Travel	:	₹7,50,000	₹4,30,000	₹3,20,000
	9.7	Attending or organizing Workshop/Seminar	:	0	0	0
	9.8	***Total Revenue expenditure*** *(9.4+9.5+9.6+9.7)*	:	₹ 74,30,000	₹ 37,70,000	₹ 36,60,000
	9.9	Contingency	:	₹ 3,71,500	₹ 1,88,500	₹ 1,83,000
	9.10	Institutional Overhead	:	₹ 15,00,000	₹ 15,00,000	0
	9.11	Applicable taxes/duties/charges etc.		0	0	0
	9.12	**Grand Total** (9.3 + 9.8 + 9.9 + 9.10 + 9.11)	: :	₹ 2,93,01,500	₹ 2,54,58,500	₹ 38,43,000

10.0 Phasing of fund requirement (in percentage) with respect to activities/milestone

11.0 Outlay for land & Building: **None**

12.0 Justification for land & building: **Not Applicable**

13.0 Outlay for Equipment:

Generic Name of equipment and accessories with major specifications		Number	Imported/ Indigenous	Estimated Cost (in Rs.)	Foreign Exchange Component
1.	Data acquisition system + NI DAQ	5	Indigenous	₹2,750,000	None
2.	Field data acquisition systems	3	Indigenous	₹750,000	
3.	High end vibration sensors for validation + Thermal camera	1	Indigenous	₹2,500,000	
4.	High Speed Camera	1	Imported	₹6,500,000	EUR 71,000
5.	Experimental setup (Motor-generator, DSO, Dynamometer)	1	Indigenous	₹2,000,000	
6.	Local Server at Singrauli	1	Indigenous	₹2,000,000	
7.	Server at IIT Kharagpur	1	Indigenous	₹3,500,000	
Total				₹20,000,000	

14.0 Justification for Equipment: Details mentioned in the body of the proposal.

15.0 Outlay for consumable materials:

Head	Outlay		
	1st Yr.	2nd Yr.	Total
Electronics components (sensors, embedded systems, IC, cables, power supplies)	₹600,000	₹600,000	₹1,200,000
Cloud service	₹200,000	0	₹200,000

AUROBINDA ROUTRAY

Aurobinda Routray
Professor, Department of Electrical Engineering, IIT Kharagpur
West Bengal, 721302
Email: aroutray@ee.iitkgp.ac.in
Phone: +91 9434041933(MOBILE), +91-3222-283072(OFFICE)

1) Educational Qualification

	Degree	Year	Subject/ Topic(s)	University/Institution
1	BTech	1989	Electrical Engineering	National Institute of Technology, Rourkela
2	MTech	1991	Electrical Engineering	Indian Institute of Technology Kanpur
3	Ph.D	1999	Design of Intelligent controllers for HVDC converters	Sambalpur University
4	Post Doctorate	2004	Mathematical Signal Processing	Purdue University

2) Past Experience in the field of research

	Positions held	Name of the Institute	From	To
I	Professor	IIT Kharagpur	23/8/2011	Now
II	Associate dean infrastructure (Electrical)	IIT Kharagpur	23/06/2020	06/01/2023
III	Associate professor	IIT Kharagpur	09/08/2004	22/08/2011
IV	Lecturer	N.I.T. Rourkela	1992	1999
V	Research Engineer	N.I.T. Rourkela	1991	1992

3) Number of research projects handled
15

4) Commercial application of research findings of other research projects handled by the investigator(s) in the past

i) Research Projects

S.No.	Client/Organisation's name	Duration of project
1	GAIL India	2021-2023
2	GEOPIC, ONGC, Dehra Dun, Uttarakhand	2018-2023
3	GEOPIC, ONGC, Dehra Dun, Uttarakhand	2017-2021
4	Department of Atomic Energy (DAE),Government of India, Board of Research in Nuclear Sciences (BRNS), BRNS Secretariat, Mumbai	2018-2020
5	Science and Engineering Research Board (SERB),Department of Science and Technology, New Delhi	2018-2021
6	MHRD , NEW DELHI & IIT BHU	2016-2019
7	INMAS, Defence Research and Development Organisation, New Delhi	2016-2018
8	Defence Institute of Physiology and Allied Sciences, DRDO	2015-2016
9	Research Designs & Standards Organisation (RDSO) Ministry of Railways,	2012-2016
10	MHRD , NEW DELHI	2014-2017
11	MHRD , NEW DELHI	2013-2016
12	Defence Institute of Physiology and Allied Sciences, DRDO	2014-2015
13	Samsung Electronics Co. Ltd	2012-2013
14	DeitY, Govt. of India	2010-2013
15	MHRD , NEW DELHI	2010-2017

ii) Consulting Experience

S.No.	Client/ Organisation's name	Duration of assignment
1	Somnioglobal (USA)	2021-2022
2	Epoq Labs Inc. Austin (USA)	2018-2019
3	Steel Authority of India Ltd. DGM and HoG (SP Group) RDCIS, Ranchi, Jharkhand	2017-2018
4	TITAN COMPANY LIMITED, Bangalore	2016-2017
5	EBIW Smartech Private Limited, STEP IIT Kharagpur	2017-2017
7	GEOPIC, ONGC, India	2016-2021
8	Air Bus India Ltd	
9	General Motors (India Science Lab)	

5) Papers published (India / Abroad) by the Investigators, etc.

S.No.	Author(s)	Title	Name of Journal	Volume	Page	Year
1.	Patitapaban Palo, Aurobinda Routray	An optimized patch-point based approach for seismic fault interpretation using CNN	Exploration Geophysics		1-17	2023
2.	Patitapaban Palo, Aurobinda Routray, Rahul Mahadik, Sanjai Singh	Fault detection in seismic data using graph convolutional network	The Journal of Supercomputing		1-29	2023
3.	Suparna Rooj, Aurobinda Routray, Manas K Mandal	Feature based analysis of thermal images for emotion recognition	Engineering Applications of Artificial Intelligence	120	105809	2023
4.	Tiash Ghosh, Ratul Kishore Saha, Sanjai Kumar Singh, Mamata Jenamani, Aurobinda Routray	LRSTV: A low-rank total variation-based seismic fault preserving denoising algorithm	Journal of Applied Geophysics		104948	2023
5.	Ishita Sar, Aurobinda Routray, Biswajit Mahanty	International Journal of Intelligent Transportation Systems Research	International Journal of Intelligent Transportation Systems Research		1-19	2023
6.	Sanhita Mishra, Sarat Chandra Swain, Tapaswini Biswal, Aurobinda Routray	Tree Based Fault Classification in Underground Cable	Procedia Computer Science	218	524-531	2023
7.	Gagandeep Singh, Rahul Mahadik, William K Mohanty, Aurobinda Routray, Deepan Datta, Sanket Smarak Panda	Seismic multi-attribute approach using visual saliency for subtle fault visualization	Exploration Geophysics		1-8	2022
8.	Arvind Kumar Gupta, Aurobinda Routray, VN Achutha Naikan	Detection of power theft in low voltage distribution systems: a review from the Indian perspective	IETE Journal of Research	68	4180-4197	2022
9.	Monorama Swain, Bubai Maji, P Kabisatpathy, Aurobinda Routray	A DCRNN-based ensemble classifier for speech emotion recognition in Odia language	Complex & Intelligent Systems	8	4237-4249	2022
10	Meghabriti Pramanik, Aurobinda Routray,	A two-stage adaptive symmetric-strong-tracking square-root	Electric Power Systems Research	210	108133	2022

	Pabitra Mitra	cubature Kalman filter for harmonics and interharmonics estimation				
11	Arpita Mondal, Aurobinda Routray, Sreeraj Puravankara	Parameter identification and co-estimation of state-of-charge of Li-ion battery in real-time on Internet-of-Things platform	Journal of Energy Storage	51	104370	2022
12	Soumi Chaki, Aurobinda Routray, William K Mohanty	A probabilistic neural network (PNN) based framework for lithology classification using seismic attributes	Journal of Applied Geophysics	199	104578	2022
13	Monorama Swain, Bubai Maji, P Kabisatpathy, Aurobinda Routray	A DCRNN-based ensemble classifier for speech emotion recognition in Odia language	Complex & Intelligent Systems		1-13	2022
14	Shubhobrata Bhattacharya, Aurobinda Routray	Heterogeneous face quality assessment	Neural Computing and Applications		1-14	2022
15	Satyajit Nayak, Aurobinda Routray, Monalisa Sarma, Satarupa Uttarkabat	GNN based Embedded Framework for Consumer Affect Recognition using Thermal Facial ROIs	IEEE Consumer Electronics Magazine			2022
16	Teja Mannepalli, Aurobinda Routray	Sparse algorithms for EEG source localization	Medical & Biological Engineering & Computing	59	2325-2352	2021
17	Dasgupta, Anirban,Routray, Aurobinda	A New Multi-resolution Analysis Method for Electrooculography Signals	IEEE Transactions on Neural Systems and Rehabilitation Engineering	1-1		2021
18	Dasgupta Anirban, Routray Aurobinda	Piecewise empirical mode Bayesian estimation – A new method to denoise electrooculograms	Biomedical Signal Processing and Control	70		2021
19	Vrinda Prajapati ,Rajlakshmi Guha,Aurobinda Routray	Multimodal prediction of trait emotional intelligence–Through affective changes measured using non-contact based physiological measures	PLoS ONE			2021
20	Ruphan Swaminathan, Sanhita Mishra, Aurobinda Routray & Sarat Chandra Swain	A CNN-LSTM-based fault classifier and locator for underground cables	Neural Computing and Applications (2021)			2021
21	Satyajit Nayak, Bingi Nagesh, Aurobinda Routray, Monalisa Sarma	A Human–Computer Interaction framework for emotion recognition through time-series thermal video sequences	Computers & Electrical Engineering	93	107280	2021
22	Rahul Mahadik, Gagandeep Singh, Aurobinda Routray	Multispectral Coherence Analysis for Better Fault Visualization in Seismic Data	IEEE Geoscience and Remote Sensing Letters			2021
23	Laxmi Shaw, Aurobinda Routray	Brain State Classification with Group l1-norm Sparse PDC as Novel Features for EEG	IEEE Sensors Journal	0	0	2021
24	Shubhobrata	Simplified Face Quality Assessment	Pattern Recognition Letters	147	108-114	2021

	Bhattacharya, Chirag Kyal, Aurobinda Routray	(SFQA)				
25	A. K. Samanta, A. Routray, S. R. Khare and A. Naha	Minimum Distance-Based Detection of Incipient Induction Motor Faults Using Rayleigh Quotient Spectrum of Conditioned Vibration Signal	IEEE Transactions on Instrumentation and Measurement	70	1-11	2021
26	Gupta, Arvind Kumar, Aurobinda Routray, and VN Achutha Naikan	Detection of Power Theft in Low Voltage Distribution Systems: A Review from the Indian Perspective	IETE Journal of Research (2020)	1	1-18	2020
27	Samanta, Anik Kumar Routray, Aurobinda Khare, Swanand R. Naha, Arunava	Direct estimation of multiple time-varying frequencies of non-stationary signals	Signal Processing	169		2020
28	Das, Samiran Routray, Aurobinda Deb, Alok Kanti	Efficient tensor decomposition approach for estimation of the number of endmembers in a hyperspectral image	Journal of Applied Remote Sensing	14	1	2020
29	Biswas, Rajashree Routray, Aurobinda	A Simple Method to Localize and Monitor the Emissive Sources in a Converter Panel	IEEE Transactions on Industrial Electronics	0	1	2020
30	Bhattacharya, Shubhobrata Dasgupta, Anirban Routray, Aurobinda	Multi-directional local adjacency descriptors (MDLAD) for heterogeneous face recognition	IET Image Processing	14	982-994	2020
31	Biswas, Rajashree Routray, Aurobinda	Application of EMR Signature in Health Assessment and Monitoring of IGBT-Based Converters	IEEE Transactions on Power Electronics	35	1899-1906	2020
32	Anirban Dasgupta; Som Madhab Bhattacharya; Aurobinda Routray	A System for Noncontact Estimation of Cognitive Load Using Saccadic Parameters Based on a Serio-Parallel Computing Framework	IEEE Transactions on Cognitive and Developmental Systems	11	450-459	2019
33	Das, Samiran Chakraborty, Sohom Routray, Aurobinda Deb, Alok Kanti	Library-aided bilinear unmixing of hyperspectral image using subspace clustering and multistep pruning	Journal of Applied Remote Sensing	13	1	2019
34	Pramanik, Meghabriti Routray, Aurobinda Mitra, Pabitra	Accurate Real-Time Estimation of Power System Transients Using Constrained Symmetric Strong Tracking Square-Root Cubature Kalman Filter	IEEE Access	7	165692-165709	2019
35	Subhasmita Sahoo, Aurobinda Routray	Dual estimation based vocal tract shape computation	International Journal of Speech Technology	22	575-584	2019
36	Soumi Chaki, Aurobinda Routray, William K Mohanty	A Novel Preprocessing Method Based on Variational Mode Decomposition for Reservoir Characterization Using Support Vector Regression	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing		1-10	2019
37	Das, Samiran, Shubhobrata Bhattacharya, Aurobinda Routray, and Alok Kani	Band Selection of Hyperspectral Image by Sparse Manifold Clustering.	IET Image Processing	13	1625-1635	2019

	Deb.					
38	Das, Samiran, Aurobinda Routray, and Alok Kanti Deb.	Sparsity Measure Based Library Aided Unmixing of Hyperspectral Image.	IET Image Processing	13	2077-2085	2019
39	Agarwal, Sumit, Harshit S. Sikchi, Suparna Rooj, Shubhobrata Bhattacharya, and Aurobinda Routray.	Illumination-Invariant Face Recognition by Fusing Thermal and Visual Images via Gradient Transfer.	In Advances in Intelligent Systems and Computing	943	658-670	2019
40	Bhattacharya, Shubhobrata, Suparna Rooj, and Aurobinda Routray.	QDF: A Face Database with Varying Quality.	Signal Processing: Image Communication	74	13-20	2019
41	Chaudhuri, Aritra, and Aurobinda Routray.	Driver Fatigue Detection Through Chaotic Entropy Analysis of Cortical Sources Obtained From Scalp EEG Signals.	IEEE Transactions on Intelligent Transportation Systems		1-14	2019
42	Bhattacharya, Shubhobrata, Gowtham Sandeep Nainala, Suparna Rooj, and Aurobinda Routray.	Local Force Pattern (LFP): Descriptor for Heterogeneous Face Recognition.	Pattern Recognition Letters	125	63-70	2019
43	Das, Samiran, and Aurobinda Routray.	Covariance Similarity Approach for Semiblind Unmixing of Hyperspectral Image.	IEEE Geoscience and Remote Sensing Letters		1-5	2019
44	Mohanty, Ramanarayan, S. L. Happy, and Aurobinda Routray.	A Semisupervised Spatial Spectral Regularized Manifold Local Scaling Cut With HGF for Dimensionality Reduction of Hyperspectral Images.	IEEE Transactions on Geoscience and Remote Sensing	57	1-13	2019
45	Dasgupta, Anirban, Som Madhab Bhattacharya, and Aurobinda Routray.	A System for Non-Contact Estimation of Cognitive Load Using Saccadic Parameters Based on a Serio-Parallel Computing Framework.	IEEE Transactions on Cognitive and Developmental Systems		1-1	2019
46	Shaw, Laxmi, and Aurobinda Routray.	A New Framework to Infer Intra- and Inter-Brain Sparse Connectivity Estimation for EEG Source Information Flow.	IEEE Sensors Journal	18	10134-44	2018
47	Mannepalli, Teja, and Aurobinda Routray.	Certainty Based Reduced Sparse Solution for Dense Array EEG Source Localization.	IEEE Transactions on Neural Systems and Rehabilitation Engineering	27		2018
48	Basu, Anushree, Anirban Dasgupta, Anirud Thyagharajan, Aurobinda Routray, Rajlakshmi Guha, and Pabitra Mitra.	A Portable Personality Recognizer Based on Affective State Classification Using Spectral Fusion of Features.	IEEE Transactions on Affective Computing			2018
49	Shaw, Laxmi, and Aurobinda Routray.	Topographical Assessment of Neurocortical Connectivity by Using Directed Transfer Function and Partial Directed Coherence during	Cognitive Processing			2018

		Meditation.				
5	Shaw, Laxmi, Daleef Rahman, and Aurobinda Routray.	Highly Efficient Compression Algorithms for Multichannel EEG.	IEEE Transactions on Neural Systems and Rehabilitation Engineering	26	957-68	2018
5	Sahoo, Subhasmita, and Aurobinda Routray.	Detecting Aggression in Voice Using Inverse Filtered Speech Features.	IEEE Transactions on Affective Computing	9	217-26	2018
5	Dasgupta, Anirban, Daleef Rahman, and Aurobinda Routray.	A Smartphone-Based Drowsiness Detection and Warning System for Automotive Drivers.	IEEE Transactions on Intelligent Transportation Systems		1-10	2018
5	Chaki, Soumi, Aurobinda Routray, and William K. Mohanty.	Well-Log and Seismic Data Integration for Reservoir Characterization: A Signal Processing and Machine-Learning Perspective.	IEEE Signal Processing Magazine	35	72-81	2018
5	Mohanty, Ramanarayan, S. L. Happy, and Aurobinda Routray.	Spatial-Spectral Regularized Local Scaling Cut for Dimensionality Reduction in Hyperspectral Image Classification.	IEEE Geoscience and Remote Sensing Letters		1-5	2018
5	Samanta, Anik Kumar, Arunava Naha, Aurobinda Routray, and Alok Kanti Deb.	Fast and Accurate Spectral Estimation for Online Detection of Partial Broken Bar in Induction Motors.	Mechanical Systems and Signal Processing	98	63-67	2018
5	Das, Samiran, Aurobinda Routray, and Alok Kanti Deb.	Fast Semi-Supervised Unmixing of Hyperspectral Image by Mutual Coherence Reduction and Recursive PCA.	Remote Sensing	10	1106	2018
5	Mohanty, Ramanarayan, S. L. Happy, and Aurobinda Routray.	A Supervised Geometry-Aware Mapping Approach for Classification of Hyperspectral Images.	IEEE Geoscience and Remote Sensing Letters	15	582-86	2018
5	Swain, Monorama, Aurobinda Routray, and P. Kabisatpathy.	Databases Features and Classifiers for Speech Emotion Recognition: A Review.	International Journal of Speech Technology	21	93-120	2018
5	Chakraborty, Suvodip, Anirban Dasgupta, and Aurobinda Routray.	Localization of Eye Saccadic Signatures in Electrooculograms Using Sparse Representations with Data Driven Dictionaries.	Pattern Recognition LetterS		5043-5048	2017
6	Happy, S. L., Priyadarshi Patnaik, Aurobinda Routray, and Rajlakshmi Guha.	The Indian Spontaneous Expression Database for Emotion Recognition.	IEEE Transactions on Affective Computing	8	131-42	2017
6	Mukherjee, Rashmi, Suman Tewary, and Aurobinda Routray.	Diagnostic and Prognostic Utility of Non-Invasive Multimodal Imaging in Chronic Wound Monitoring: A Systematic Review.	Journal of Medical Systems	41	46	2017
6	Sengupta, Anwesha, Anirban Dasgupta, Aritra Chaudhuri, Anjith George, Aurobinda Routray, and Rajlakshmi	A Multimodal System for Assessing Alertness Levels Due to Cognitive Loading.	IEEE Transactions on Neural Systems and Rehabilitation Engineering	25	1037-46	2017

	Guha.					
63	Shaw, Laxmi, Aurobinda Routray, and Sirin Sanchay.	A Robust Motifs Based Artifacts Removal Technique from EEG.	Biomedical Physics & Engineering Express	3	35010	2017
64	Happy, S. L., and Aurobinda Routray.	Fuzzy Histogram of Optical Flow Orientations for Micro-Expression Recognition.	IEEE Transactions on Affective Computing		1-1	2017
65	Naha, Arunava, Anik Kumar Samanta, Aurobinda Routray, and Alok Kanti Deb.	Low Complexity Motor Current Signature Analysis Using Sub-Nyquist Strategy with Reduced Data Length.	IEEE Transactions on Instrumentation and Measurement	66	3249-59	2017
66	Naha, Arunava, Korada Ravisri Thammayyabbabu, Anik Kumar Samanta, Aurobinda Routray, and Alok Kanti Deb.	Mobile Application to Detect Induction Motor Faults.	IEEE Embedded Systems Letters	9	117-20	2017
67	Chaki, Soumi, Aurobinda Routray, and William K. Mohanty.	A Diffusion Filter Based Scheme to Denoise Seismic Attributes and Improve Predicted Porosity Volume.	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing	10	5265-74	2017
68	Dasgupta, Anirban, Suvodip Chakraborty, and Aurobinda Routray.	A Two-Stage Framework for Denoising Electrooculography Signals.	Biomedical Signal Processing and Control	31	231-37	2017
69	Mukherjee, Ayan, Aurobinda Routray, and Anik Kumar Samanta.	Method for Online Detection of Arcing in Low-Voltage Distribution Systems.	IEEE Transactions on Power Delivery	32	1244-52	2017
70	George, Anjith, and Aurobinda Routray.	A Score Level Fusion Method for Eye Movement Biometrics.	Pattern Recognition Letters	82	207-15	2016
71	Basu, Anushree, Aurobinda Routray, Rashmi Mukherjee, and Suprosanna Shit.	Infrared Imaging Based Hyperventilation Monitoring through Respiration Rate Estimation.	Infrared Physics and Technology	77	382-90	2016
72	Sahoo, Subhasmita, and Aurobinda Routray.	A Novel Method of Glottal Inverse Filtering.	IEEE/ACM Transactions on Audio Speech and Language Processing	24	1230-41	2016
73	George, Anjith, and Aurobinda Routray.	Fast and Accurate Algorithm for Eye Localization for Gaze Tracking in Low Resolution Images.	IET Computer Vision	10	660-69	2016
74	Naha, Arunava, Anik Kumar Samanta, Aurobinda Routray, and Alok Kanti Deb.	A Method for Detecting Half-Broken Rotor Bar in Lightly Loaded Induction Motors Using Current.	IEEE Transactions on Instrumentation and Measurement	65	1614-25	2016
75	Sahadevan, Anand S., Aurobinda Routray, Bhabani S. Das, and Saquib Ahmad.	Hyperspectral Image Preprocessing with Bilateral Filter for Improving the Classification Accuracy of Support Vector Machines.	Journal of Applied Remote Sensing	10	25004	2016
76	Pradhan, Tapan, Bibek Kabi, Ramanarayan	Development of Numerical Linear Algebra Algorithms in Dynamic	International Journal of Circuit Theory and	44	1222-62	2016

	Mohanty, and Aurobinda Routray.	Fixed-Point Format: A Case Study of Lanczos Tridiagonalization.	Applications			
77	Sengupta, Anwesha, Subhadeep Datta, Sibsambhu Kar, and Aurobinda Routray.	EEG Synchronization and Brain Networks.	International Journal of Biomedical and Clinical Engineering	4	1-11	2015
78	Happy, S. L., and Aurobinda Routray.	Automatic Facial Expression Recognition Using Features of Salient Facial Patches.	IEEE Transactions on Affective Computing	6	1-12	2015
79	Sengupta, Anwesha, Sibsambhu Kar, and Aurobinda Routray.	Study of Loss of Alertness and Driver Fatigue Using Visibility Graph Synchronization.	Innovative Research in Attention Modeling and Computer Vision Applications		171-93	2015
80	Swain, Monorama, Subhasmita Sahoo, Aurobinda Routray, P. Kabisatpathy, and Jogendra N. Kundu.	Study of Feature Combination Using HMM and SVM for Multilingual Odiya Speech Emotion Recognition.	International Journal of Speech Technology	18	387-93	2015
81	Chaki, Soumi, Aurobinda Routray, and William K. Mohanty.	A Novel Preprocessing Scheme to Improve the Prediction of Sand Fraction from Seismic Attributes Using Neural Networks.	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing	8	1808-20	2015
82	Verma, Akhilesh K., Aurobinda Routray, and William K. Mohanty.	Assessment of Similarity between Well Logs Using Synchronization Measures.	IEEE Geoscience and Remote Sensing Letters	11	2032-36	2014
83	Naha, Arunava, Anik Kumar Samanta, Aurobinda Routray, and Alok Kanti Deb.	Determining Autocorrelation Matrix Size and Sampling Frequency for MUSIC Algorithm.	IEEE Signal Processing Letters	22	1016-20	2014
84	Chaki, Soumi, Akhilesh K. Verma, Aurobinda Routray, William K. Mohanty, and Mamata Jenamani.	Well Tops Guided Prediction of Reservoir Properties Using Modular Neural Network Concept: A Case Study from Western Onshore India.	Journal of Petroleum Science and Engineering	123	155-63	2014
85	Verma, Akhilesh K., Soumi Chaki, Aurobinda Routray, William K. Mohanty, and Mamata Jenamani.	Quantification of Sand Fraction from Seismic Attributes Using Neuro-Fuzzy Approach.	Journal of Applied Geophysics	111	141-55	2014
86	Dasgupta, Anirban, Anjith George, S. L. Happy, and Aurobinda Routray.	A Vision-Based System for Monitoring the Loss of Attention in Automotive Drivers.	IEEE Transactions on Intelligent Transportation Systems	14	1825-38	2013
87	Pradhan, Tapan, Bibek Kabi, Aurobinda Routray, and Gonnabhaktula Anirudh.	Fixed-Point Hestenes SVD Algorithm for Computing Eigen Faces.	International Journal of Circuits, Systems and Signal Processing	7	312-21	2013
88	Kar, Sibsambhu, and Aurobinda Routray.	Effect of Sleep Deprivation on Functional Connectivity of EEG Channels.	IEEE Transactions on Systems, Man, and Cybernetics Part	43	666-72	2013

			A:Systems and Humans			
89	Dasgupta, Anirban, Anjith George, S. L. Happy, Aurobinda Routray, and Tara Shanker.	An On-Board Vision Based System for Drowsiness Detection in Automotive Drivers.	International Journal of Advances in Engineering Sciences and Applied Mathematics	5	94-103	2013
90	Ghosh, Shrabani, Anirban Krishna Bhattacharyya, Siddhartha Mukhopadhyay, and Aurobinda Routray.	Improved Estimation of Kinematic State of Reentry Ballistic Targets with Modeling of Colored and Nonstationary Seeker Noise.	IEEE Transactions on Aerospace and Electronic Systems	48	3186-3206	2012
91	Mohanty, Mihir Narayan, Aurobinda Routray, Ashok Kumar Pradhan, and Prithviraj Kabisatpathy.	Power Quality Disturbances Classification Using Support Vector Machines with Optimised Time-Frequency Kernels.	International Journal of Power Electronics	4	181	2012
92	Ghosh, Shrabani, Siddhartha Mukhopadhyay, and Aurobinda Routray.	New State and Measurement Models for Endo-Atmospheric Tracking of Ballistic Targets Using Seeker Measurements.	IEEE Transactions on Aerospace and Electronic Systems	48	1192-1209	2012
93	Kar, Sibsambhu, Aurobinda Routray, and Bibhukalyan Prasad Nayak.	Functional Network Changes Associated with Sleep Deprivation and Fatigue during Simulated Driving: Validation Using Blood Biomarkers.	Clinical Neurophysiology	122	966-74	2011
94	Mohanty, Mihir Narayan, Aurobinda Routray, and Prithviraj Kabisatpathy.	Optimisation of Features Using Evolutionary Algorithm for EEG Signal Classification.	International Journal of Computational Vision and Robotics	1	297	2011
95	Senapati, Kedarnath, and Aurobinda Routray.	Comparison of ICA and WT with S-Transform Based Method for Removal of Ocular Artifact from EEG Signals.	Journal of Biomedical Science and Engineering	4	341-51	2011
96	Senapati, Kedarnath, West Bengal, and William Kumar Mohanty.	Blind Deconvolution of Seismic Signals in Time and Frequency Domain : A Study on 2004 Sumatra Earthquake.	International Journal of Signal Processing, Image Processing and Pattern Recognition	4	29-47	2011
97	Patil, Dinesh, Niva Das, and Aurobinda Routray.	Implementation of Fast-ICA: A Performance Based Comparison between Floating Point and Fixed Point DSP Platform.	Measurement Science Review	11	118-24	2011
98	Gupta, Supratim, Aurobinda Routray, and Anirban Mukherjee.	A New Method for Edge Extraction in Images Using Local Form Factors.	International Journal of Computer Applications	21	15-22	2011
99	Jenamani, Mamata, Aurobinda Routray, and Vikash Singh.	A Procedure Using Support Vector Data Description and Mutual Information for End Price Assessment in Online C2C Auction.	Electronic Commerce Research		321-40	2011
10	Kar, Sibsambhu, Mayank Bhagat, and Aurobinda	EEG Signal Analysis for the Assessment and Quantification of	Transportation Research Part F:	13	297-306	2010

	Routray.	Driver's Fatigue.	Traffic Psychology and Behavior			
10	Mohanty, Mihir Narayan, Anurag Kumar, Aurobinda Routray, and Prithviraj Kabisatpathy.	Evolutionary Algorithm Based Optimization for Power Quality Disturbances Classification Using Support Vector Machines.	International Journal of Control, Automation and Systems		1306-12	2010
10	Mohanty, Mihir Narayan, Vinay Kumar, Aurobinda Routray, and Prithviraj Kabisatpathy.	Classification of Power Quality Disturbances Using Parzen Kernels.	International Journal of Emerging Electric Power Systems	11		2010
10	Mohanty, Mihir Narayan, Aurobinda Routray, and Prithviraj Kabisatpathy.	Voice Detection Using Statistical Method.	Int. J. Engg. Techsci 2 (1): 120–24.	1	120-24	2010
10	Mohanty, Mihir, Aurobinda Mishra, and Aurobinda Routray.	A Non-Rigid Motion Estimation Algorithm for Yawn Detection in Human Drivers.	International Journal of Computational Vision and Robotics	1	89	2009
10	Kandali, Aditya Bihar, Aurobinda Routray, and Tapan Kumar Basu.	Cross-Lingual Vocal Emotion Recognition in Five Native Languages of Assam Using Eigenvalue Decomposition.	Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)	5909	519-24	2009
10	Das, Niva, Aurobinda Routray, and Pradipta Kishor Dash.	A Robust H ∞ Learning Approach to Blind Separation of Slowly Time-Varying Mixture of Acoustic Electromechanical Signals.	Mechanical Systems and Signal Processing	23	2049-58	2009
10	Mohanty, S. R., A. K. Pradhan, and A. Routray.	A Cumulative Sum-Based Fault Detector for Power System Relaying Application.	IEEE Transactions on Power Delivery	23	79-86	2008
10	Routray, Aurobinda, and Smarak Swain.	Fast Algorithms for Designing Variable FIR Notch Filters.	Numerical Linear Algebra with Applications	14	673-94	2007
10	Mishra, Chandrabhanu, Aurobinda Routray, and Siddhartha Mukhopadhyay.	A Computationally Efficient Winding Function Method for Calculation of Inductances in an Asymmetric Induction Motor.	Electric Power Components and Systems	35	43-61	2007
11	Mohanty, William Kumar, Aurobinda Routray, and Sankar Kumar Nath.	A New Strategy for Phase Detection in Seismic Signals Using an Adaptive Markov Amplitude Model.	Current Science	93	54-60	2007
11	Das, Niva, A Routray, and PK Dash.	ICA Methods for Blind Source Separation of Instantaneous Mixtures: A Case Study.	Neural Information Process. Letters	11	225-46	2007
11	Pradhan, A. K., A. Routray, and S. Madhan Gudipalli.	Fault Direction Estimation in Radial Distribution System Using Phase Change in Sequence Current.	IEEE Transactions on Power Delivery	22	2065-71	2007
11	Pradhan, A. K., A. Routray, and Banaja Mohanty.	Maximum Efficiency of Flexible AC Transmission Systems.	International Journal of Electrical Power and Energy Systems	28	581-88	2006
11	Pradhan, A. K., A.	A Moving Sum Approach for Fault	Electric Power	34	385-99	2006

	Routray, and S. R. Mohanty.	Detection of Power Systems.	Components and Systems			
1	Das, Debi Prasad, Swagat Ranjan Mohapatra, Aurobinda Routray, and T. K. Basu.	Filtered-s LMS Algorithm for Multichannel Active Control of Nonlinear Noise Processes.	IEEE Transactions on Audio, Speech and Language Processing	14	1875-80	2006
1	Pradhan, A. K., S. K. Meher, and A. Routray.	Communication Channel Equalization Using Wavelet Network.	Digital Signal Processing: A Review Journal	16	445-52	2006
1	Pradhan, A. K., A. Routray, and Abir Basak.	Power System Frequency Estimation Using Least Mean Square Technique.	IEEE Transactions on Power Delivery	20	1812-16	2005
1	Pradhan, Ashok Kumar, and Aurobinda Routray.	Applying Distance Relay for Voltage Sag Source Detection.	IEEE Transactions on Power Delivery	20	529-31	2005
1	Pradhan, A. K., A. Routray, S. Pati, and D. K. Pradhan.	Wavelet Fuzzy Combined Approach for Fault Classification of a Series-Compensated Transmission Line.	IEEE Transactions on Power Delivery	19	1612-18	2004
1	Pradhan, A. K., A. Routray, and B. Biswal.	Higher Order Statistics-Fuzzy Integrated Scheme for Fault Classification of a Series-Compensated Transmission Line.	IEEE Transactions on Power Delivery	19	891-93	2004
1	Routray, Aurobinda, Ashok Kumar Pradhan, and K. Prahallad Rao.	A Novel Kalman Filter for Frequency Estimation of Distorted Signals in Power Systems.	IEEE Transactions on Instrumentation and Measurement	51	469-79	2002
1	Dash, P. K., Aurobinda Routray, and Sukumar Mishra.	Neural Network Based Feedback Linearizing Controller for HVDC Links.	Electric Power Systems Research	50	125-32	1999
1	Dash, P. K., A. Routray, and A. C. Liew.	Design of an Energy Function Based Fuzzy Tuning Controller for HVDC Links.	International Journal of Electrical Power and Energy System	21	337-47	1999
1	Dash, P.K., D.P. Swain, A. Routray, and A.C. Liew.	An Adaptive Neural Network Approach for the Estimation of Power System Frequency.	Electric Power Systems Research	41	203-10	1997
1	Routray, Aurobinda, P. K. Dash, and Sanjeev K. Panda.	A Fuzzy Self-Tuning PI Controller for HVDC Links.	IEEE Transactions on Power Electronics	11	669-79	1996
1	Dash, P. K., A. C. Liew, and A. Routray.	Design of Robust Controllers for HVDC Links in AC-DC Power Systems.	Electric Power Systems Research	33	201-9	1995
1	Dash, P. K., A. Routray, and A. Mahapatra.	Design of a Variable Structure Current Regulator for Weak AC/DC Systems.	Electric Power Systems Research	27	49-59	1993

Mamata Jenamani,

Professor
Department of Industrial and Systems Engineering
Indian Institute of Technology, Kharagpur,
Phone 91-3222-283740 **(o)** 91-3222-283073 (r)
mj@iem.iitkgp,ac.in

Address
A - 7,
Indian Institute of Technology, Kharagpur,
West Bengal, India
PIN 721302

Research Interest

- E- Business, Information Systems
- Supply chain management and automation
- Decision support and analytics

Education

- 2005 PhD, Department of Industrial and Systems Engineering, IIT Kharagpur, India
- 1997 Master, Department of Computer Science and Engineering, NIT Rourkela, India (Then REC Rourkela, India)

Employment

- Professor, Department of Industrial and Systems Engineering, IIT, Kharagpur, 2018-till date
- Associate Professor, Department of Industrial and Systems Engineering, IIT, Kharagpur, 2010-2018
- Assistant Professor, Department of Industrial and Systems Engineering, IIT, Kharagpur, 2005-2010
- Research Associate, Raid Lab, Computer Science Department, **Purdue University, USA** (2003 –2004), Mentor: Prof. Bharat Bhargava
- PhD Scholar, Department of Industrial Engineering and Management, **Indian Institute of Technology, Kharagpur,** 2000-2003, Mentor: Prof. P K J Mohapatra, Prof. Sujoy Ghose
- Senior lecturer the Department of Computer Science Engineering and applications in **National Institute of Technology, Rourkela** from 1998 - 2003.
- Lecture in the Department of Computer Science Engineering and applications in **National Institute of Technology, Rourkela** from 1991 -1997.
- Software Engineer in **ORG,** 1990

Awards and Recognitions

- Winner of **Emerald/EFMD Outstanding Doctoral Research Awards** in the category of Enterprise Applications of Internet Technology.
- **DST Women Scientist Award**

Projects

Sponsored Research

1. Automated negotiation and trust management in electronic market places (2005, ISIRD, SRIC, IIT Kharagpur, 3.00 Lakhs) PI: Mamata Jenamani
2. Exploring Trust, Fraud and Privacy issues in E-Business (2005, 8 Lacks), PI: Mamata Jenamani
3. Virtual Lab for Simulation and Gaming, (2012, MHRD, 52.00 Lakhs) PI: B. Mahanty.
4. Intelligent Decision Support in Online Auction (CSIR, 2014, 11.84 Lakhs), PI: Mamata Jenamani
5. Information and Communication Technology and Waste Control in Logistics (MHRD, 2014-17, 55 Lakhs), PI: Mamata Jenamani
6. Urban Governance: Sustainable Policies and Implementations (MHRD, 2014-18, 38 Lakhs) PI: Mamata Jenamani
7. Assessment of the convergence activities under Mahatma Gandhi NREGA in Odisha (National Institute of Rural Development and Panchayati Raj, 2016-17, 12.42 Lakhs) PI: N C Nayak, Co-PI: Mamata Jenamani
8. Application of sparse layer inversion and compressive sensing for thin layer reservoir delineation using seismic data (ONGC, Value:56 Lakhs) PI: A. Routray, Co-PI: W.K. Mohanty, Mamata Jenamani.
9. E-Business Centre of Excellence (MHRD, 2014-, 4 Crores), PI: Mamata Jenamani, M K Tiwari

10. Real time monitoring and embedded analytics for quality traceability on cold chains for sea export of agricultural produce (imprint-2, SERB, DST, Value: Rs92.1 Lakhs) PI: Mamata Jenamani, Co-PI: A. Routray, J Mitra
11. Dynamic distribution planning under risk during epidemic outbreak in humanitarian supply chain, Sponsore, Coordination Committee of RAKSHAK,IIT Jodhpur (10 Lakhs)
12. Fault mapping and facies prediction using big data analytics (2018-23), (GEOPIC, ONGC , 2.16 crores)
13. A cloud based ai enabled iot platform for monitoring, diagnosis and prognosis of faults in electric motors and drives, GAIL, 1.25 crores

Consultancy Projects

14. Restructuring of OLIMMS and providing failure safe system (Client: Neyveli Lignite Corporation Limited, Value: 87.52 Lakhs; Period: 24 months) PI: R.N. Banerjee; Co-PIs: Mamata Jenamani.
15. Support for database administratorfor OLIMMS (Client: Neyveli Lignite Corporation Limited, Value: 19.42 Lakhs; Period: 36 months), PI: R.N. Banerjee; Co-PIs: Mamata Jenamani.
16. Modification of E-Proc. Software to comply with Interoperability Guidelines for Digital Signature Certificate for Neyveli Lignite Corporation Limited (Client NLC, Value Rs13 Lakhs) PI: R.N. Banerjee. Co-PIs: Mamata Jenamani.
17. To Study, design, develop, install, test, commission, implement the Web based Reverse Auction Software in NLC Ltd for Neyveli Lignite Corporation Limited (Client NLC, Value Rs. 27 Lakhs) PI Mamata Jenamani
18. Reservoir Characterization using Artificial Intelligent Techniques (Soft Computing) (Client, GEOPIC, ONGC, Value:45 Lakhs) PI: W.K. Mohanty, A. Routray, Mamata Jenamani.
19. Evaluation of remote village electrification programme, roof top solar system and bio gas (Client: Planning & Co ordination Department, Bhubaneswar, Govt. of Odisha, 16 Lakh), PI: N C Nayak, Co-PI: Mamata Jenamani

Papers published (India / Abroad) by the Investigators, etc.

- Publication list: https://scholar.google.co.in/citations?hl=en&user=9m9sxGcAAAAJ&view_op=list_works
- Publications in last 5 years

Sl	Authors	Title	Publication	vol	num	Year
1	Pattanaik Swati; Jenamani Mamata;	Identifying the cooling heterogeneity and quality decay of Indian mangoes during cold chain export by multiphysics modeling	Journal of Food Process Engineering	46	3	2023
2	Ghosh Tiash; Saha Ratul Kishore; Singh Sanjai Kumar; Jenamani Mamata; Routray Aurobinda;	LRSTV: A low-rank total variation-based seismic fault preserving denoising algorithm	Journal of Applied Geophysics	210		2023
3	Singh Amit; Thakkar Jitesh; Jenamani Mamata;	An integrated Grey-DEMATEL approach for evaluating ICT adoption barriers in manufacturing SMEs: analysing Indian MSMEs	Journal of Enterprise Information Management	35	6	2022
4	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar;	An approximate marginal spread computation approach for the budgeted influence maximization with delay	Computing	104	3	2022
5	Chetan TG; Jenamani Mamata; Sarmah SP;	Iterative Multi-Attribute Procurement Auction with Decision Support for Bid Formulation	Asia-Pacific Journal of Operational Research	39	3	2022
6	Singh Amit; Jenamani Mamata; Thakkar Jitesh J; Rana Nripendra P;	Quantifying the effect of eWOM embedded consumer perceptions on sales: An integrated aspect-level sentiment analysis and panel data modeling approach	Journal of Business Research	138		2022
7	Bardhan Subhadeep; Jenamani Mamata; Routray Aurobinda;	Improving IoT Sensor Data Quality using Kalman Filter: The case of Cold Chain monitoring for Indian Mangoes	2022 IEEE 7th International conference for Convergence in Technology (I2CT)			2022
8	Bagchi Sourav; Jenamani Mamata; Routray	Multivariate real-time Missing Value Imputation using Adaptive VAR with IoT data from a	2022 IEEE 7th International conference for Convergence in			2022

	Aurobinda;	Refrigerated Container	Technology (I2CT)			
9	Dash Ritwik; Mawa Shubham; Jenamani Mamata;	Outlier Sensitive Online Change Detection using Convex Combination of Adaptive Filters in Sensor Data Streams	2022 IEEE 7th International conference for Convergence in Technology (I2CT)			2022
10	Nayak Nikesh; Pant Pushpesh; Sarmah Sarada Prasad; Jenamani Mamata; Sinha Deepankar;	A novel Index-based quantification approach for port performance measurement: A case from Indian major ports	Maritime Policy & Management			2022
11	Saha Ratul Kishore; Pradhan Ashutosh; Ghosh Tiash; Jenamani Mamata; Singh Sanjai Kumar; Routray Aurobinda;	GPU Accelerated Parallel Implementation of Linear Programming Algorithms	Information Integration and Web Intelligence: 24th International Conference, iiWAS 2022, Virtual Event, November 28–30, 2022, Proceedings			2022
12	Saha Ratul Kishore; Ghosh Tiash; Singh Sanjai Kumar; Jenamani Mamata; Routray Aurobinda; Mondal Arpita;	Fast and Parallel Semblance Algorithm for Detecting Faults in Large Seismic Volumes	IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society			2022
13	Ghosh Tiash; Saha Ratul Kishore; Singh Sanjai Kumar; Jenamani Mamata; Routray Aurobinda;	Divergence of covariance matrices as a discontinuity attribute for 2D seismic fault detection	2022 IEEE 19th India Council International Conference (INDICON)			2022
14	Swain Satyajit; Jenamani Mamata;	Cluster analysis for identification of temperature breaks in a reefer container from IoT data	2022 International Conference on Futuristic Technologies (INCOFT)			2022
15	Singh Sandeep Kumar; Jenamani Mamata;	Cassandra-based data repository design for food supply chain traceability	VINE Journal of Information and Knowledge Management Systems	51	2	2021
16	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar;	Earned benefit maximization in social networks under budget constraint	Expert Systems with Applications	169		2021
17	Banerjee Suman; Banjare Pratik; Pal Bithika; Jenamani Mamata;	A multistep priority-based ranking for top-N recommendation using social and tag information	Journal of Ambient Intelligence and Humanized Computing	12		2021
18	Singh Sandeep Kumar; Jenamani Mamata; Dasgupta Diptiman; Das Suman;	A conceptual model for Indian public distribution system using consortium blockchain with on-chain and off-chain trusted data	Information Technology for Development	27	3	2021
19	Mitra Satanik; Jenamani Mamata;	An Approach to Identify Provocative and Problematic Content with Social Nociceptor	Proceedings of the 3rd ACM India Joint International Conference on Data Science & Management of Data (8th ACM IKDD CODS & 26th COMAD)			2021
20	Mitra Satanik; Jenamani Mamata;	Helpfulness of online consumer reviews: A multi-perspective approach	Information Processing & Management	58	3	2021
21	Singh Amit; Jenamani Mamata; Thakkar Jitesh J; Rana Nripendra P;	Propagation of online consumer perceived negativity: Quantifying the effect of supply chain underperformance on passenger car sales	Journal of Business Research	132		2021
22	Biswal Arun Kumar;	Product subsidy and expected consumption with	Computers &	15		2021

	Jenamani Mamata; Kumar Sri Krishna;	inventory inaccuracy: Implications of RFID adoption in Indian public distribution system	Industrial Engineering	9		
23	Singh Amit; Jenamani Mamata; Thakkar Jitesh; Dwivedi Yogesh K;	A Text Analytics Framework for Performance Assessment and Weakness Detection From Online Reviews	Journal of Global Information Management (JGIM)	30	8	2021
24	Narayanan Abhishek; Jenamani Mamata; Mahanty Biswajit;	Determinants of sustainability and prosperity in Indian cities	Habitat International	118		2021
25	Nayak Narayan Chandra; Sahoo Bimal Kishore; Jenamani Mamata; Mohanty Alok Ranjan; Chatterjee Runa Sen;	Does Convergence of Rural Development Schemes Improve Household Welfare? An Investigation of Mahatma Gandhi National Rural Employment Guarantee Act in Odisha, India	The Indian Journal of Labour Economics	64		2021
26	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar;	A survey on influence maximization in a social network	Knowledge and Information Systems	62		2020
27	Biswal Arun Kumar; Jenamani Mamata; Kumar Sri Krishna;	The impact of RFID adoption on donor subsidy through for-profit and not-for-profit newsvendor: Implications for Indian Public Distribution system	Socio-Economic Planning Sciences	69		2020
28	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar; Sirohi Abhinav;	A Priority-Based Ranking Approach for Maximizing the Earned Benefit in an Incentivized Social Network	Intelligent Systems Design and Applications: 18th International Conference on Intelligent Systems Design and Applications (ISDA 2018) held in Vellore, India, December 6-8, 2018, Volume 1			2020
29	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar;	Maximizing the earned benefit in an incentivized social networking environment: a community-based approach	Journal of Ambient Intelligence and Humanized Computing	11		2020
30	Pal Bithika; Jenamani Mamata;	Personalized Ranking in Collaborative Filtering: Exploiting l-th Order Transitive Relations of Social Ties	Proceedings of the 7th ACM IKDD CoDS and 25th COMAD			2020
31	Mitra Satanik; Jenamani Mamata;	A Method to Estimate Perceived Quality and Perceived Value of Brands to Make Purchase Decision Using Aspect-Based Sentiment Analysis	Innovations in Computer Science and Engineering: Proceedings of 7th ICICSE			2020
32	Singh Amit; Jenamani Mamata; Thakkar Jitesh;	Do online consumer reviews help to evaluate the performance of automobile manufacturers?	Journal of Enterprise Information Management			2020
33	Mitra Satanik; Jenamani Mamata;	OBIM: A computational model to estimate brand image from online consumer review	Journal of Business Research	114		2020
34	Mitra Satanik; Jenamani Mamata;	Hybrid improved document-level embedding (hide)	arXiv preprint arXiv:2006.01203			2020
35	Swain Satyajit; Jenamani Mamata; Routray Aurobinda; KumarSingh Sanjai; Tondon Rajeev; Bahuguna C;	PySLI: A software for Sparse Layer Inversion using Python	Proceedings of the 13th Biennial International Concerence and Exhibition, Kochi, Japan			2020
36	Banerjee Suman; Pal Bithika; Jenamani Mamata;	Budgeted influence maximization with tags in social networks	Web Information Systems Engineering–WISE 2020: 21st International Conference, Amsterdam, The Netherlands,			2020

			October 20–24, 2020, Proceedings, Part I 21			
37	Banerjee Suman; Pal Bithika; Jenamani Mamata;	DySky: dynamic skyline queries on uncertain graphs	Web Information Systems Engineering–WISE 2020: 21st International Conference, Amsterdam, The Netherlands, October 20–24, 2020, Proceedings, Part I 21			2020
38	Bardhan Subhadeep; Bagchi Sourav; Jenamani Mamata; Routray Aurobinda;	Non-Invasive method using Contact-less Sensors and Embedded Platform for Monitoring Quality determining factors of Indian Mangoes	IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society			2020
39	Pattanaik Swati; Jenamani Mamata;	Numerical analysis of cooling characteristics of Indian mangoes using digital twin	IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society			2020
40	Pal Bithika; Jenamani Mamata;	Trust inference using implicit influence and projected user network for item recommendation	Journal of Intelligent Information Systems	52		2019
41	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar;	Maximizing the earned benefit in an incentivized social networking environment: An integer programming-based approach	Proceedings of the ACM India Joint International Conference on Data Science and Management of Data			2019
42	Banerjee Suman; Jenamani Mamata; Pratihar Dilip Kumar;	ComBIM: A community-based solution approach for the Budgeted Influence Maximization Problem	Expert Systems with Applications	125		2019
43	Dey Girish Ch; Jenamani Mamata;	Optimizing fortification plan of capacitated facilities with maximum distance limits	Opsearch	56	1	2019
44	Dey Atanu; Jenamani Mamata; Thakkar Jitesh J;	Cross-D-vectorizers: a set of feature-spaces for cross-domain sentiment analysis from consumer review	Multimedia Tools and Applications	78		2019
45	Ramkumar M; Schoenherr Tobias; Wagner Stephan M; Jenamani Mamata;	Q-TAM: A quality technology acceptance model for predicting organizational buyers' continuance intentions for e-procurement services	International Journal of Production Economics	216		2019
46	Singh Sandeep Kumar; Jenamani Mamata; Garg Chetan; Alirajpurwala Hasan;	Multi-echelon supply network analysis with interactive visualization	2019 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon)			2019
47	Biswal Arun Kumar; Jenamani Mamata; Kumar Sri Krishna;	Warehouse efficiency improvement using RFID in a humanitarian supply chain: Implications for Indian food security system	Transportation Research Part E: Logistics and Transportation Review	109		2018
48	Dey Atanu; Jenamani Mamata; Thakkar Jitesh J;	Senti-N-Gram: An n-gram lexicon for sentiment analysis	Expert Systems with Applications	103		2018
49	Chetan TG; Jenamani Mamata; Sarmah Sarada Prasad;	Two-stage multi-attribute auction mechanism for price discovery and winner determination	IEEE Transactions on Engineering Management	66	1	2018
50	Biswal Arun Kumar; Jenamani Mamata;	Leveraging ICT for food security: An analysis in the context of PDS in India	Social Transformation–Digital Way: 52nd Annual Convention of the Computer Society of India, CSI 2017, Kolkata, India, January 19-21, 2018, Revised			2018

			Selected Papers 52			
51	Singh Sandeep Kumar; Biswal Arun Kumar; Kumar Sonu; Singh Tanya; Jenamani Mamata;	Optimal equipment positioning and technology selection for RFID enabled warehouses	Social Transformation–Digital Way: 52nd Annual Convention of the Computer Society of India, CSI 2017, Kolkata, India, January 19-21, 2018, Revised Selected Papers 52			2018
52	Pal Bithika; Banerjee Suman; Jenamani Mamata;	Threshold-based heuristics for trust inference in a social network	2018 Eleventh International Conference on Contemporary Computing (IC3)			2018
53	Pal Bithika; Jenamani Mamata;	Kernelized probabilistic matrix factorization for collaborative filtering: exploiting projected user and item graph	Proceedings of the 12th ACM Conference on Recommender Systems			2018
54	Kumar Vinay; Jenamani Mamata;	Identification and Prioritization of urban issues from Smart City data	2018 International Conference on Smart City and Emerging Technology (ICSCET)			2018
55	Mitra Satanik; Jenamani Mamata;	SentiCon: A Concept Based Feature Set for Sentiment Analysis	2018 IEEE 13th International Conference on Industrial and Information Systems (ICIIS)			2018
56	Dey Atanu; Jenamani Mamata; Thakkar Jitesh J;	Senti-N-Gram: An n-gram lexicon for sentiment analysis (vol 103, pg 92, 2018)	EXPERT SYSTEMS WITH APPLICATIONS	108		2018

Sunita Mishra, Ph.D.

Assistant Professor

Department of Mining Engineering
Indian Institute of Technology Kharagpur
West Bengal – 721302
Primary Email: smishra@mining.iitkgp.ac.in
Alternate Email: sunita.mishra256@gmail.com
Phone: +91-3222-214824 (O), +91-9717253877 (M)
Website: https://www.sunitamishra.com/



Research Expertise

Micromechanical Modelling, Size Effect and Scale Effect in Brittle Materials: Rock, Concrete, Ceramics; Experimental Techniques under Dynamic Loading, Development of Constitutive Relations, Rock Dynamics and Fracture Propagation in Rocks, Artificial Intelligence in Rock Dynamics, Blast and Ballistic Loading, Machine Vibration and Isolators, Rock and Tool Interaction.

Academic Background

Indian Institute of Technology (IIT) Delhi, India
Doctor of Philosophy (Ph.D.) **CGPA: 8.5/10.0** **June 2014 – Feb 2019**
Thesis: Characterization of High Strain Rate Response of Rock and Its Application in Blast Analysis of Tunnels.

Indian Institute of Technology (IIT-ISM) Dhanbad, India
Masters in Technology (M.Tech.) **CGPA: 8.89/10.0** **June 2012 – May 2014**
Specialization in Tunnelling and Underground Space Technology **(Gold Medalist)**
Thesis: Designing of Ventilation System in Underground Metro Railway System.

Biju Patnaik University of Technology (BPUT), Odisha, India
Bachelors in Technology (B.Tech.) **CGPA: 8.58/10.0** **Jul 2008 – May 2012**
Specialization in Mechanical Engineering **(Gold Medalist)**
Thesis: Comparison between Effectiveness of Various Fin Shapes.

Ramadevi Womens College, Bhubaneswar, India
12th Board **MARK: 78.0 %** **Apr 2008**

DAV Public School MCL Kalinga Area, Talcher, India
10th Board **MARK: 89.6 %** **Apr 2006**

Professional Experiences

Assistant Professor **Feb 2020 – Continuing**
Department of Mining Engineering
Indian Institute of Technology Kharagpur, India

Post-doctoral Research Fellow **July 2019 – Jan 2020**
SIMLab, Centre for Advanced Structural Analysis
NTNU, Norway

IUSSTF Intern Fellow **Sept 2018 – Jan 2019**
HEMI - Hopkins Extreme Materials Institute
Johns Hopkins University, USA

Visiting Ph.D. Student **May 2018 – July 2018**
SIMLab, Centre for Advanced Structural Analysis
NTNU, Norway

Teaching Experiences

Assistant Professor at IIT Kharagpur, India

1) Rock and Tool Interaction	(MI60210)	(New Course Introduced)
2) Mining Machinery and Automation	(MI31203)	
3) Mining Machinery Laboratory	(MI39203)	
4) Engineering Drawing and Computer Graphics	(CE13003)	
5) Rock Dynamics	(MI60064/MI60218)	
6) Bulk Material Handling	(MI62001)	
7) Drilling Technology	(MI60032)	
8) Mine Surveying Laboratory	(MI29202/MI29002)	
9) Safety Engineering Laboratory	(MI49001)	
10) Safety Health and Environment Laboratory IV	(MI69010)	
11) Planning and Design Laboratory	(MI49201)	

Institute Administrative Experiences

1. Core Committee Member of Induction Program 2024.
2. Member of the Dias Management Committee in the 69th Convocation 2023 of IIT Kharagpur.
3. Assistant Warden of Sir Ashutosh Mukherjee (SAM) Guest House and Bachelor Flat-1 from April 2023-Present.
4. Organizer of International Women's Day 2023 and 2024, IIT Kharagpur.
5. Member of the Dias Management Committee in the 68th Convocation 2022 of IIT Kharagpur.
6. Faculty mentor for the Society for Mining, Metallurgy & Exploration: IIT Kharagpur Local Chapter.
7. Organizer of International Women's Day 2022, IIT Kharagpur going to be held from 6th-8th March 2022.
8. Member of the IoE Review Committee.
9. Member of Central Research Facility, IIT Kharagpur.
10. Associated Member of Center of Excellence - Safety Engineering and Analytics (CoE-SEA).
11. Assistant Warden of Sarojini Naidu/Indira Gandhi Hall of Residence 2021-2023.
12. Executive Member of Kharagpur Local Centre, Institution of Engineers (India) for the session 2021-2013
13. Joint Secretary of International Women's Day 2021, IIT Kharagpur held from 6th-8th March 2021.
14. Member of Gender Environment Committee, IIT Kharagpur.

Departmental Administrative Experiences

1. In charge of the Mining Machinery Laboratory (2020-Till Present).
2. Member of Building Infrastructure Committee in the Department of Mining Engineering, IIT Kharagpur (2023-Present).
3. Member of the Faculty Search Committee, Department of Mining Engineering, IIT Kharagpur (2020-23).
4. Faculty advisor to Dual Degree students of the batch with 2020 entry (2020-Till Present).
5. Departmental Time Table In-Charge (2020-Till Present).
6. Departmental ERP In-Charge (2020-Till Present).
7. Member of Departmental Purchase Committee (2020-Till Present).

Research Laboratory and Facility Developed

1. Material Dynamics Research Laboratory in the Department of Mining Engineering, IIT Kharagpur.
2. Designed and developed a 48 mm diameter coupled static and dynamic loading split Hopkinson pressure bar device capable of both compressive and tensile loading conditions in the Department of Mining Engineering, IIT Kharagpur.
3. Designed and developedp a novel CNC Rock Cutting facility in Mining Machinery Laboratory of Mining Engineering Department at IIT Kharagpur.

Software Proficiencies

Programming Language	:	FORTRAN, MATLAB
CAD/CAM Software	:	SOLIDWORKS, AUTOCAD, PRO_E
FEM Software	:	ABAQUS, LS-DYNA, Rocscience Module

Editorial Board Member

1. Guest Editor in MGMI Transaction Journals for the upcoming special issue of SMART 2020 on the theme "Safety and Occupational Health".

Reviewer of International Journal and Conferences

1. International Journal of Impact Engineering
2. International Journal of Rock Mechanics and Mining Sciences
3. Rock Mechanics and Rock Engineering
4. Journal of Materials in Civil Engineering
5. Engineering Fracture Mechanics
6. Shock and Vibration
7. Arabian Journal of Geosciences
8. Brazilian Society of Mechanical Sciences and Engineering
9. Advances in Civil Engineering
10. Geomechanics and Engineering, An International Journal
11. Materials Today Communications
12. Bulletin of Engineering Geology and the Environment
13. Construction and Building Materials
14. Indian Geotechnical Journal
15. Journal of Building Engineering

Awards and Achievements

1. Session Co-Chair in Session 2 of the International Conference on Industrial Engineering and Analytics (ICONIEA 2024) organized by the Department of Industrial and Systems Engineering (ISE), Indian Institute of Technology (IIT) Kharagpur from 12th -16th February 2024.
2. Session Co-Chair in Session 1 and Session 3 of the International Conference on Safety, Health and Analytics-Driven Governance for Sustainable Development" (SHADG2024) organized by Center of Excellence - Safety Engineering and Analytics (CoE-SEA), IIT Kharagpur.
3. SERB International Travel Grant for participating in ISRM Congress 2023 held from 9-14 October 2023 in Salzburg, Austria.
4. Nominated by INAE, India for Poster Presentation at National Frontiers of Engineering 2023 Conclave under the aegis of INAE- SERB conclave on "Atmanirbhar Technologies –Engineering Secured Future" hosted by BIT, Mesra, Ranchi.
5. Institute of Engineers India (IEI) Young Engineers Award 2021-22 in Mining Engineering Discipline.
6. Session Co-Chair in Session 1 of Track 3 of the International Conference on Maintenance and Intelligent Asset Management 2021 (ICMIAM2021) held from 12-15 December 2021.
7. Nominated for Erasmus+ ICM KA 107 to Universidad Santiago de Compostela.
8. Visiting Ph.D. student to HEMI Laboratory, Johns Hopkins University, Baltimore, U.S.A. - Selected through Indo-U.S. Women Overseas Student Internship in Science, Technology, Engineering, Mathematics and Medicine (WISTEMM) Program 2018 Call Awards - September 2018 to January 2019, Advisor: Prof. K.T. Ramesh.
9. Visiting Ph.D. student to Centre for Advanced Structural Analysis (CASA), NTNU-Norwegian University of Science and Technology (NTNU), Norway - May 2018 to July 2018, Advisor: Prof. Tore Borvik and Prof. Magnus Langseth.
10. DST International Travel Grant for participating in RocDyn-3, An International Conference on Rock Dynamics and Applications held on 25-29 June 2018 in NTNU, Trondheim, Norway.
11. ICEM 2018 Conference Attendance Award at the 18th International Conference on Experimental Mechanics going to be held in Brussels, Belgium from 1-5 July 2018.
12. Best poster award (Defence domain) in Industry day 2017 in Indian Institute of Technology (IIT) Delhi.
13. CSIR Travel Grant for participating in EUROCK 2017, An International ISRM Conference held on 20-22 June in Ostrava-Poruba, Czech Republic.
14. Best paper award in ARM9 - 9th Asian Rock Mechanics Symposium "Harmonizing Rock Mechanics with Sustainable Economic Development", 18-20 October 2016, Bali, Indonesia.
15. Honoured with Gold Medal from Indian Institute of Technology (IIT-ISM) Dhanbad for Best Tunnelling and Underground Space Technology Graduate of the year - 2014.

Memberships

1. Individual Membership of International Society of Rock Mechanics (ISRM)
2. Associate Member of American Society of Civil Engineers (ASCE)
3. Individual Member of International Society of Impact Engineering
4. Lifetime Associate Member of Institute of Engineers, India.

Sponsored Research Projects

1. **Project Title** Thermo-Mechanical Characterization of Rock Under Dynamic Loading Conditions
   Funding Agency Board of Research in Nuclear Sciences, Department of Science and Technology India.
   Amount and Duration **INR 47,62,600/-** **Oct 2023 - Sept 2026 (Ongoing)**

2. **Project Title** Development of an unconventional flapping-based sediment-aeration system for the brackish water aquaculture
   Funding Agency Department of Science and Technology (DST), India
   Amount and Duration **INR 48,66,440/-** **Sept 2021 – Sept 2024 (Ongoing)**

3. **Project Title** Development of Split Hopkinson Pressure Bar Device to Assess Fracture Propagation in Notched Rock Specimen.
   Funding Agency Institute Scheme for Innovative Research and Development (ISIRD), IIT Kharagpur India.
   Amount and Duration **INR 42,60,169/-** **June 2021 – June 2024 (Completed)**

4. **Project Title** A Study on the Dynamic Tensile Response of Rocks and its Application in Blast Analysis of Subsurface Structures.
   Funding Agency Start-up Research Grant (SRG), Science and Engineering Research Board, India
   Amount and Duration **INR 29,97,500/-** **Jan 2021 – Dec 2022 (Completed)**

List of Invited Talks Delivered

1. Invited Seminar on "High Strain Rate Characterisation of Rocks and its Application in Blast Analysis of Tunnels." at National Yang Ming Chiao Tung University, Taiwan on 11th December 2023.
2. Invited Seminar on "High Strain Rate Characterisation of Rocks and its Application in Blast Analysis of Tunnels." at National Metallurgical Laboratory, Jamshedpur on 30th May 2023.
3. Invited Seminar on "High Strain Rate Characterisation of Rocks and its Application in Blast Analysis of Tunnels." at IIT Kharagpur-NYCU, Taiwan Lecture Series on 10th February 2023.
4. Guest Lecture on "High Strain Rate Characterisation of Rocks and its Application in Blast Analysis of Tunnels." at Norwegian University of Science and Technology, Norway on 13th June 2018.
5. Invited Lecture on "Experimental Stress Analysis and Crashworthiness of Vehicle." at Maruti Suzuki Limited, Gurgaon, India.

List of Invited Members of Eminent Academic/Regulatory Bodies

1. Member of the International Society of Rock Mechanics (ISRM) Commission of Rock Dynamics.
2. Member of Diamond Core and Water Well Drilling Sectional Committee in the Bureau of Indian Standards (BIS).
3. President of Asset Management Society of India, Kharagpur Local Chapter.
4. Judge in Regional Level Science, Mathematics and Environment Exhibition for Children 2022-23 and 50th Rashtriaya Bal Vaigyanik Pradarshini 2023 at KV IIT Kharagpur.
5. Board of Studies Member of the Mechanical Engineering Department at Gandhi Institute of Technological Advancements, Bhubaneswar.

List of Workshops Organised

1. Coordinator of 1st Webinar series on "Systems and Practices during a Pandemic: Experiences and Learning's from COVID 2nd Wave" organized by Asset Management Society India.
2. Co-Convener and Organiser of Post-SMART2020 Webinar Workshop on Safety in Mining and SMART Technology.

List of Conferences Organised

1. Co-convener of the International Conference on Geo-Resorces Engineering, Energy and Nature (GREEN) 2025 organized by the Department of Mining Engineering, IIT Kharagpur (Date: 8-10th December 2025).
2. Organising Secretary of the International Conference on Safety, Health and Analytics-Driven Governance for Sustainable Development" (SHADG2024) organized by Center of Excellence - Safety Engineering and Analytics (CoE-SEA), IIT Kharagpur held from 29-30 January 2024.
3. Co-Convener and Organizing secretary of "International Conference on Maintenance and Intelligent Asset Management 2021 (ICMIAM2021)" going to be held from 12-15 December 2021.
4. Secretary of an International Webinar "SMART 2020" held from 16-18th December 2020.

List of Journals Published

1. Chakraborty, T., Mishra, S., Loukus, J., Halonen, B., Bekkala, B. (2016). "Characterization of Three Himalayan Rocks using a Split Hopkinson Pressure Bar." International Journal of Rock Mechanics and Mining Sciences, 85, 112-118. (SCI and SCOPUS Indexed, Impact Factor: 7.135)
2. Mishra, S., Meena, H., Parashar, V., Khetwal, A., Chakraborty, T., Matsagar, V., Chandel, P., Singh, M. (2018). "High Strain Rate Response of Rocks under Dynamic Loading Using Split Hopkinson Pressure Bar." Geotechnical and Geological Engineering, Springer, 36(1), 531-549, https://doi.org/10.1007/s10706-017-0345-2. (SCOPUS Indexed)
3. Mishra, S., Chakraborty, T., Matsagar, V., Loukus, J., Bekkala, B. (2018). "High Strain-Rate Characterization of Deccan Trap Rocks Using SHPB Device." Journal of Materials in Civil Engineering, ASCE, 30(5), 04018059-1-9, DOI: 10.1061/(ASCE)MT.1943-5533.0002229. (SCI and SCOPUS Indexed, Impact Factor: 2.33)
4. Chaudhary, R., Mishra, S., Chakraborty, T., Matsagar, V. (2018). "Vulnerability Analysis of Tunnel Linings Under Blast Loading." International Journal of Protective Structures. DOI: 10.1177/2041419618789438 (SCOPUS Indexed)
5. Mishra, S., Khetwal, A., Chakraborty, T. (2019). "Dynamic Characterisation of Gneiss." Rock Mechanics and Rock Engineering, Springer, 52(1), 61-81. (SCI and SCOPUS Indexed, Impact Factor: 7.381)
6. Mishra, S., Chakraborty, T. (2019). "Determination of High Strain Rate Stress-Strain Response of Granite for Blast Analysis of Tunnels." Journal of Engineering Mechanics, ASCE, 145(8), 04019057-1-23. DOI: 10.1061/(ASCE)EM.1943-7889.0001627 (SCI and SCOPUS Indexed, Impact Factor: 2.66)
7. Mishra, S., Chakraborty, T., Basu, D., Lam, N. (2019). "Characterization of Sandstone for Application in Blast Analysis of Tunnel." Geotechnical Testing Journal, ASTM, 43. DOI: 10.1520/GTJ20180270. (SCI and SCOPUS Indexed, Impact Factor: 1.843)
8. Mishra, S., Khetwal, A., Chakraborty, T. (2019). "Physio-Mechanical Characterisation of Rocks." Journal of Testing and Evaluation, ASTM, 49(3), DOI: 10.1520/JTE20180955. (SCI and SCOPUS Indexed, Impact Factor: 1.264)
9. Mishra, S., Chakraborty, T., Rao, K.S. (2020). "Dynamic Response of Two Extrusive Igneous Rocks using Split Hopkinson Pressure Bar Test." Journal of Materials in Civil Engineering, ASCE, In Press, DOI: 10.1061/(ASCE)MT.1943-5533.0003725. (SCI and SCOPUS Indexed, Impact Factor: 2.33)
10. Mishra, S., Yadav, H., Chakraborty, T., Kapuria, S. (2022). "Physio-Mechanical Characterisation of Limestone and Dolomite for its Application in Blast Analysis of Tunnels" Journal of Engineering Mechanics, ASCE, 148(5), DOI: 10.1061/(ASCE)EM.1943-7889.0002100. (SCI and SCOPUS Indexed, Impact Factor: 2.66)
11. Mishra, S., Chakraborty, T. (2022). "Physical and Mechanical Characterisation of Metadolerite." Geotechnical and Geological Engineering, Springer, 1-12. (SCOPUS Indexed)
12. Mishra, S., Khetwal, A., Chakraborty, T., Basu, D. (2022). "Effect of Loading Characteristics and Specimen Size in Split Hopkinson Pressure Bar Test on High-Rate Behavior of Phyllite." Archives of Civil and Mechanical Engineering, Accepted. (SCI and SCOPUS Indexed, Impact Factor: 4.369)
13. Roy, R., Chakraborty, S., Bisai, R., Pal, S.K., Mishra, S. (2023). "Gravity Blind Backfilling of Abandoned Underground Mine Voids Using Suitable Mix Proportion of Fill Materials and Method of Filling." Geotechnical and Geological Engineering, Springer, https://doi.org/10.1007/s10706-022-02371-8. (SCOPUS Indexed)

List of Conferences Published

1. Mishra, S., Parashar, V., Chakraborty, T., Matsagar, V., Chandel, P., Mangla, V., Singh, M. (2015). “High Strain Rate Response of Sedimentary Rocks under Dynamic Loading using Split Hopkinson Pressure Bar.” Proceedings of PROTECT2015 Fifth International Workshop on Performance, Protection & Strengthening of Structures under Extreme Loading, East Lansing, MI, US. 29-30 June, 5, 88-96. Presented.
2. Mishra, S., Chakraborty, T., Basu, D. (2016). “High Strain Rate Response of Soils - A Review.” Japanese Geotechnical Special Publication, 3(2), 80-85. Presented. (Journal)
3. Anuradha, Mishra, S., Chakraborty, T., Matsagar, V., Chandel, P., Singh M. (2016). “High Strain Rate Response of Himalayan Quartzitic Gneiss by using Split Hopkinson Pressure Bar.” 2nd International Conference on Rock Dynamics and Applications (RocDyn-2), 18-20 May, Suzhou, China. Presented. (Scopus Indexed)
4. Meena, H., Mishra, S., Chakraborty, T., Matsagar, V., Chandel, P., Mangla, V., Singh, M. (2016). “Dynamic Characterization of Rocks Subjected to Intermediate and High Strain Rates.” 2016 ISRM International Symposium, Rock Mechanics and Rock Engineering: From the Past to the Future, 29-31 August, Cappadocia, Turkey. Presented. (Scopus Indexed)
5. Meena, H., Mishra, S., Chakraborty, T., Matsagar, V., Chandel, P., Mangla, V., Singh, M. (2016). “High Strain Rate Characterization of Himalayan Limestone.” 2016 ISRM Regional Symposium, ARMS9 - 9th Asian Rock Mechanics Symposium "Harmonizing Rock Mechanics with Sustainable Economic Development", 18-20 October, Bali, Indonesia. Presented. (Best Paper Award)
6. Mishra, S., Meena, H., Chakraborty, T., Chandel, P., Singh, M. (2017). “High Strain Rate Characterization of Himalayan Dolomite.” 11th International Symposium on Plasticity and Impact Mechanics, IMPLAST 2016, Procedia Engineering, 173, 822-829. Presented. (Scopus Indexed)
7. Mishra, S., Chakraborty, T., Matsagar, V. (2017). “Dynamic Characterization of Himalayan Quartzite using SHPB.” Eurock 2017, the ISRM European Rock Mechanics Symposium, 20-22 June, Ostrava-Poruba, Czech Republic. Presented. (Scopus Indexed)
8. Mishra, S., Chakraborty, T., Matsagar, V., Basu, D. (2018). “High Strain Rate Characterization of Himalayan Limestone using SHPB.” 3rd International Conference on Rock Dynamics and Applications (RocDyn-3), 25-29 June, NTNU, Trondheim, Norway. Presented. (Scopus Indexed)
9. Mishra, S., Chakraborty, T. (2018). “Mechanical Characterization of Himalayan Dolomite.” 5th GeoChina International Conference, Civil Infrastructures Confronting Severe Weathers and Climate Changes: From Failure to Sustainability, 23-25 July, HangZhou, China. (Scopus Indexed)
10. Mishra, S., Chakraborty, T., Matsagar, V. (2018). “Mechanical Characterization of Compact Basalt by using SHPB device.” ICEM 2018 - 18th European International Conference on Experimental Mechanics, 1-5 July, Brussels, Belgium. Presented.
11. Mishra, S., Chakraborty, T., Matsagar, V. (2018). “Strain Rate Effect on the Mechanical Properties of Granite.” Engineering Mechanics Institute Conference 2018, ASCE, 29 May - 1 June, M.I.T., Cambridge, MA.
12. Mishra, S., Chakraborty, T. (2018). “Mechanical Characterization of Rock and its Application in Blast Analysis of Tunnels.” Indian Geotechnical Conference 2018, Frontiers and New Horizons in Geotechnical Engineering, Springer, December 13-15, 2018, IISc Bangalore, India. Presented.
13. Ray, K., Mishra, S. (2022). “Numerical Study on the Effect of Shockwaves used for Percussive Drilling.” ARMS12 - 12th Asian Rock Mechanics Symposium - an ISRM Regional Symposium, Hanoi, November 22-26, 2022.
14. Pinapatruni, V. R. G., Ranjan, R., Charan, D., Mishra, S., Dash, S. M. (2022). “Characterisation of the New Open Surface Recirculating Water Tunnel Facility at the Indian Institute of Technology Kharagpur.” The 9th International and 49th National Conference of Fluid Mechanics and Fluid Power (FMFP-2022), 14-16 December, IIT Roorkee, India. Presented. Springer Proceedings (Scopus Index)
15. Ranjan, M., Mishra, S., Pal, S.K., Venkateswarlu, G., Sridhar, K., Kanakaiah, M., Subhani, S. M. (2022). “Determination of Barrier Pillars Stability of an Abandoned Underground Mine to an Open Cast Mine: A Numerical Case Study.” 32nd National Convention of Mining Engineers, Institute of Engineer's India, Durgapur Local Chapter, 3-4 December 2022. Presented.
16. Gangavarapu, T., Kumari, A., Samal, R.K., Mishra, S. (2023). “Effect of Dynamic Compressive Loading Condition on Pre-Defined Crack Propagation.” 6th International Conference on Protective Structures, Auburn, May 2023.
17. Mishra, S., Chakraborty, T., Basu, D. (2023). “Effect of Specimen Size and Loading Rate on Dynamic Properties of Sandstone Rock.” 6th International Conference on Protective Structures, Auburn, May 2023.
18. Samal, R.K., Gangavarapu, T., Mishra, S. (2023). “Behaviour of a Pre-defined Crack Propagating under Dynamic Tensile Loading Condition”, 15th International ISRM Congress 2023 & 72nd Geomechanics Colloquium Challenges in Rock Mechanics and Rock Engineering, October 09 - 14, 2023, Presented.
19. Mohapatra, A., Kumari, A., Mishra, S. (2023). “Thermo-mechanical Properties of Indian Sandstone under Dynamic Compressive Loading Condition.”, 15th International ISRM Congress 2023 & 72nd

Geomechanics Colloquium Challenges in Rock Mechanics and Rock Engineering, October 09 - 14, 2023, Presented.

20. Mohapatra, A., Mishra, S., Chen, W., Arora, K. (2023). “Geothermal Gradient of Underground Structure Subjected to Blast Loads.”, 18th Conference of the Associated Research Centers for the Urban Underground Space (ACUUS 2023 Singapore), November 01 - 04, 2023. Paper Accepted.
21. Samal, R.K., Mishra, S. (2023). “Micromechanics Study on the Failure Behaviour of Rock under Dynamic Loading Conditions”, National Frontiers of Engineering 2023 Conclave, INAE, BIT, Mesra, Ranchi, June 24-26, 2023, Presented.
22. Mukhopadhyay, A., Charan, D. Dash, S.M., Mishra, S. (2023). “Investigation of Frequency Effects on the Self-Propelled Heaving Airfoils”, Proceedings of the 10th International and 50th National Conference on Fluid Mechanics and Fluid Power (FMFP), December 20-22, 2023, IIT Jodhpur, India. Presented.
23. Samal, R.K., Mishra, S. (2024). “Dynamic Characterization of Synthetic Sandstone Rock under the Influence of Self-Induced Crack.”, Eurock 2024 - New Challenges in Rock Mechanics and Rock Engineering - an ISRM Regional Symposium, Alicante, Spain, July 15-19, 2024, Paper Accepted.
24. Samal, R.K., Mishra, S. (2024). “New Rock Mass Classification based on Fracture Propagation in Rock under Dynamic Loading Conditions.”, 1st International Rock Mass Classification Conference (RMCC), Oslo, Norway, October 30-31, 2024, Paper Accepted.
25. Samal, R.K., Mishra, S. (2024). “Calibration of Split Hopkinson Pressure Bar and Strain Gauges Used for Dynamic Characterization of Rocks”, 13th Asian Rock Mechanics Symposium (ARMS13), 22-27th September 2024, New Delhi. Paper Accepted.
26. Mohapatra, A., Mishra, S. (2024). “Stability Analysis of Different Tunnel Shapes and Sizes under Static and Dynamic Loading Conditions.”, 13th Asian Rock Mechanics Symposium (ARMS13), 22-27th September 2024, New Delhi. Paper Accepted.
27. Biswal, A., Samal, R.K., Mishra, S., Gorai, A.K. (2024). “Numerical Study on the Failure Behaviour of Rocks Using Mohr-Coulomb and Drucker-Prager Failure Criteria.”, 13th Asian Rock Mechanics Symposium (ARMS13), 22-27th September 2024, New Delhi. Paper Accepted.
28. Biswal, A., Samal, R.K., Mishra, S. (2024). “XFEM Analysis on the Crack Propagation in Different Rock Types under Quasi-static Loading Condition.”, 4th International Conference on Coupled Processes in Fractured Geological Media: Observation, Modeling, and Application (CouFrac2024) to be held in Kyoto, Japan, November 13-15, 2024, Abstract Accepted.

Sunil Manohar Dash, Ph.D.

Assistant Professor

Department of Aerospace Engineering
Indian Institute of Technology Kharagpur
West Bengal – 721302
Primary Email: smdash@aero.iitkgp.ac.in
Alternate Email: sunilmanohardash@gmail.com
Phone: +91-3222-304512 (O), +91-9958673652 (M)
Website: https://www.sunilmanohardash.com/



Research Expertise

Computational Fluid Dynamics; Lattice Boltzmann Method; Immersed Boundary Method; Experimental Flow Measurement; Particle Image Velocimetry (PIV) Visualisation; MAV-UAV-Flapping Airfoil Dynamics; Complex Cavity-Natural Convection; Particle Sedimentation; Active Flow Separation Control.

Academic Background

National University of Singapore, Singapore
Doctor of Philosophy (Ph.D.) — CGPA: 4.58/5.0 — Aug 2010 – Dec 2014
Thesis: Development of a flexible forcing immersed boundary-lattice Boltzmann method and its applications in thermal and particulate flows.

National Institute of Technology, Rourkela, India
Bachelor in Technology (B.Tech.) — CGPA: 9.34/10.0 — Jul 2005 – Apr 2009
Thesis: Study of cryogenic cycles with ASPEN-HYSIS simulation

Govt. Junior College, Rourkela, India
12th Board — MARK: 88.0 % — Apr 2004

Chinmaya Vidyalaya, Rourkela, India
10th Board — MARK: 89.0 % — Apr 2002

Professional Experiences

Assistant Professor — Apr 2018 – Continuing
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

Assistant Professor — Dec 2017 – Apr 2018
Department of Mechanical Engineering
Indian Institute of Technology (ISM) Dhanbad, India

Post-doctoral Research Fellow — Jan 2017 – Dec 2017
International Design Centre (IDC)
Singapore University of Technology and Design, Singapore

Post-doctoral Research Fellow — Jan 2015 – Dec 2016
Department of Mechanical Engineering
National University of Singapore, Singapore

Research Assistance — Apr 2014 – Dec 2014
Department of Mechanical Engineering
National University of Singapore, Singapore

Teaching Experiences

Assistant Professor at IIT Kharagpur, India
1) Introduction to Aerodynamics (AE21001)

2) Physics of Fluid Flow Experiments (AE40037/AE60037)
3) Industrial Aerodynamics (AE51018/AE60006)
4) Mechanics (ME10001)
5) Engineering Drawing and CAD (CE13003/CE13001)
6) Aerodynamics Laboratories - I (AE29002)
7) Seminar - I (AE69001)

Assistant Professor at IIT (ISM) Dhanbad, India

1) Internal Combustion Engine (MMC16103)
2) Engineering Graphics (MMC11101)

Graduate Tutor (Teaching Assistant) at NUS, Singapore

1) Introduction to Fluid Mechanics I (ME2134E)
2) Introduction to Fluid Mechanics II (ME2135E)

Graduate Lab Assistant at NUS, Singapore

1) Stability of Floating Bodies (ME2134)
2) Flow past a NACA Aerofoil (ME2135E)
3) Drag on a settling Sphere (ME2134E)
4) Characteristic of Centrifugal Pump (ME2135E)

Administrative Experiences

Assistant Warden of LBS Hall — Jan 2022 – Dec 2023
Indian Institute of Technology Kharagpur, India

Member of Departmental Purchase Committee — Jan 2022 – Jun 2023
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

Co-In-Charge of Aeromodelling laboratory — Jul 2019 – Jun 2023
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

Co-In-Charge of Boeing Student Project — Jul 2019 – Jun 2023
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

Member of Departmental Academic Committee — Jan 2018 – Jul 2022
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

NCC Care Taker for 1 Bengal EME Coy — Mar 2019 – Mar 2021
Indian Institute of Technology Kharagpur, India

Faculty Advisor (UG and DD) — Jul 2019 – Jul 2024
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

In-Charge of Departmental library — Jul 2018 – Jul 2019
Department of Aerospace Engineering
Indian Institute of Technology Kharagpur, India

Research Laboratory and Facility Developed

1) Developed a Bio-inspired Aero-Hydrodynamics Research Laboratory in the Department of Aerospace Engineering, IIT Kharagpur.
2) Developed a recirculating water tunnel experimental facility in the Department of Aerospace Engineering, IIT Kharagpur.
3) Developed a novel flapping mechanism to mimic the flight of the insect and bird in the Department of Aerospace Engineering, IIT Kharagpur.

4) Developed a robotic jellyfish, an alternate underwater vehicle in the Department of Aerospace Engineering, IIT Kharagpur.

Software Proficiencies

Programming Language	:	FORTRAN, MATLAB
CAD/CAM Software	:	CATIA, AUTOCAD, INVENTOR, SOLIDWORKS
CFD Software	:	FLUENT, GAMBIT, STAR CCM, COMSOL
PIV Software	:	FLOW MANAGER, DAVIS

Reviewer of International Journal and Conferences

1) Journal of Fluid Mechanics (JFM)
2) Physics of Fluids (POF)
3) Computers and Fluids (CAF)
4) Ocean Engineering (OE)
5) Journal of Fluids and Structure (JFS)
6) International Journal of Heat and Mass Transfer (IJHMT)
7) Numerical Heat Transfer (NHT)
8) Heat and Mass Transfer (HAMT)
9) Journal of Heat Transfer (JHT)
10) Journal of Thermal Science and Engineering Applications (TSEA)
11) International Communication in Heat and Mass Transfer (ICHMT)
12) International Journal of Micro Air Vehicles (IJMAV)
13) Journal of Applied Fluid Mechanics (JAFM)
14) Energy Reports (ER)
15) Computers and Mathematics with Applications (CMA)
16) Iranian Journal of Science and Technology Transactions A: Science (IJSCTT)
17) International/National Conference on Fluid Mechanics and Fluid Power (FMFP)
18) International Conference on Theoretical, Applied, Computational and Experimental Mechanics (ICTACEM)
19) International Conference on Recent Advances in Sustainable Energy Research (RAISER)
20) International Conference on Ship and Offshore Technology (ICSOT)

Awards and Achievements

1) Received Faculty Excellence Award Year 2021-22 in the Assistant Professor Level at IIT Kharagpur.
2) Received Young Engineers Award in the Aerospace Engineering Discipline from the Institutions of Engineers India (IEI) for the Year 2020-21.
3) Received Odisha Young Scientists Award in the Aerospace Engineering Discipline from Odisha Bigyan Academy, Department of Science and Technology Odisha for the Year 2019-20.
4) Received International Travel Support (ITS) from the Department of Science and Technology India to attend 72nd APS DFD Conference in USA in the year 2019. (Declined)
5) Received National University of Singapore (NUS) Ph.D. Research Fellowship for the years 2010-2014.
6) Received Best Conference Paper award in the Fluid Mechanics session in the conference IC-RIDME-2018, Shillong, Meghalaya, India.
7) Chaired conference sessions on Aerodynamics Study in ICAFM-2016, Malaysia, on Active Flow Control in 70th DFD meeting, APS-2017, Denver, USA and on Fluid Mechanics in IC-RIDME-2018, Shillong, Meghalaya, India.
8) Honoured with a Gold Medal from Institution of Engineers India (IEI) for being the Best Mechanical Engineering Graduate of the year – 2009.
9) Honoured with an Institute Silver Medal from National Institute of Technology, Rourkela for being the Best Mechanical Engineering Graduate of the year – 2009.
10) Received *Summer Research Fellowship* – year 2008, from Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR).
11) Ranked 1st in 4th, 5th, 6th, 7th and 8th semesters of my B.Tech. studies. Secured SGPA 10.0/10.0 in 5th and 7th semesters of my B.Tech. studies.

Memberships

1) Professional member of APS (American Physical Society)
2) Professional member of IEEE (Institute of Electrical and Electronics Engineers)

3) Lifetime member of IEI (Institutions of Engineers India)

Sponsored Research Projects

1) **Project Title** Development of an unconventional flapping-based sediment-aeration system for the brackish water aquaculture
   Funding Agency Department of Science and Technology (DST), India
   Amount and Duration **INR 49,00,000 /-** **Year 2021 (Ongoing)**

2) **Project Title** A study on the propulsive performance of the flapping hydrofoil near the ground
   Funding Agency ISIRD, IIT Kharagpur, India
   Amount and Duration **INR 28,00,000 /-** **Year 2019 (Ongoing)**

3) **Project Title** A study on the aerodynamic and propulsion performance of tubercle flapping airfoils
   Funding Agency Aeronautical Research & Development Board (AR&DB), India
   Amount and Duration **INR 59,00,000 /-** **Year 2019 (Ongoing)**

4) **Project Title** An investigation of the unconventional tandem flapping foil propulsion mechanism for UWV
   Funding Agency Science and Engineering Research Board (SERB), India
   Amount and Duration **INR 46,00,000 /-** **Year 2019 -2022 (Completed)**

5) **Project Title** Building a water tunnel research facility in the Department of Aerospace Engineering, IIT Kharagpur.
   Funding Agency Diamond Jubilee Grant, IIT Kharagpur, India
   Amount and Duration **INR 30,00,000 /-** **Year 2018 (Completed)**

List of Invited Talks Delivered

1) Aerodynamics of Flapping Airfoils, 2nd Novemeber 2022, at Institutions of Engineers India (IEI) Kharagpur Local Chapter monthly lecture series.
2) Investigation of Propulsive Performance of Two-dimensional Flapping Airfoils, 1st March 2021, in the Short Term Training Program on "Applied Computational Fluid Dynamics for Automotive, Space and Defence Sector" in Mechanical Engineering Department, A D Patel Institute of Technology, India.
3) Lattice Boltzmann and Immersed Boundary Method for Fluid-Solid Interactions, 12th December 2019, in the Mechanical Engineering Department, NCTU Taiwan.
4) A study of high frequency flapping aerodynamics, 13th December 2018, in the Mechanical Engineering Department, NCTU Taiwan.

List of Short-Term Courses Organised

1) "Flow Visualisation and Measurement Techniques in the Incompressible and Compressible Flows", 01st – 05th August 2022, AE-Seminar room, IIT Kharagpur. (Self-Sponsored, 40 Participants).
2) "CFD and Experiments on Fluid-Structure Interactions for both Rigid and Elastic Materials", 14th – 18th October 2019, AE-Seminar room, IIT Kharagpur. (TEQIP-Sponsored, 30 Participants).
3) "Experimental Flow Visualisation and Measurement Techniques for both Incompressible and Compressible Flow Regimes", 12th – 15th March 2018, AE-Seminar room, IIT Kharagpur. (Self-Sponsored, 50 Participants).

List of Seminars Organised

1) Talk by Prof. Lua Kim Boon from Department of Mechanical Engineering, NYCU Taiwan on " Flapping Wing Micro-Aerial Robot", 05th August 2022, at AE-Seminar room, IIT Kharagpur.
2) Talk by Dr. Kishora Shetty from Boeing Research & Technology, India on "Reimagining the Aerospace Materials & Their Advancements for 21st Century ",10th March 2022, AE Department Webinar, IIT Kharagpur.
3) Talk by Mr. Partha Adhikari from Boeing Research & Technology, India on "Aircraft Predictive Maintenance: Trends, Opportunities and Challenges",07th December 2021, AE Department Webinar, IIT Kharagpur.
4) Talk by Prof. Cheng Sheng Huang from Department of Mechanical Engineering, NCTU Taiwan on "Applications of gradient grating period guided-mode resonance filter", 27th November 2018, at AE-Seminar room, IIT Kharagpur.
5) Talk by Prof. Tsung Lin Chen from Department of Mechanical Engineering, NCTU Taiwan on "MEMS Logic Gate", 27th November 2018, at AE-Seminar room, IIT Kharagpur.

6) Talk by Prof. Lua Kim Boon from Department of Mechanical Engineering, NCTU Taiwan on "Experimental work on insect aerodynamics and flight", 15th March 2018, at AE-Seminar room, IIT Kharagpur.

List of Conferences Organised

1) In the Organising Secretariat of "International Conference on Theoretical Applied Computational and Experimental Mechanics, (ICTACEM-2021)" 20th – 22nd December 2021, Indian Institute of Technology Kharagpur, India.
2) In the International Advisory Committee of "International Conference on Recent Advances in Sustainable Energy Research, (RAISER Conference-2021)" 6th – 7th July 2021, University of Science Malaysia, Penang, Malaysia.
3) In the Co-Organising Committee of "International Conference on Recent Advances in Sustainable Energy Research, (RAISER Conference-2019)" 16th December 2019, University of Science Malaysia, Penang, Malaysia.

List of Media Coverage

1) Pablo Valdivia y Alvarado, Kenneth Tracy, Christine Yogiaman, Sunil Manohar Dash, Pamela Dychengbeng Chua. (2017). "Patterned Flow: Augmenting Air Movement in Urban Environment." ARCHIFEST 2017 (Archi-Interfaces Exhibition); The URA Centre City Gallery 3F City Canvas, 45 Maxwell Road, Singapore. http://archifest.sg/2017/archi-interfaces/

List of Journals Published

1. A Numerical Study on the Drag Reduction and Wake Regime Control of the Tandem Circular Cylinders using Splitter Plates by by Sikdar P., **Dash S. M.**, Sinhamahapatra K. P., Journal of Computational Science (Accepted)
2. Mixed Convection in a Lid Driven Square Cavity using Lattice Boltzmann Method: Effects of Thermal Gradient Direction and Moving Lid Length by Bhunia A., **Dash S. M.**, Numerical Heat Transfer, Part B: Fundamentals, (Accepted)
3. Influence of the Pivot Location on the Thrust and Propulsive Efficiency Performance of a Two-dimensional Flapping Elliptic Airfoil in a Forward Flight by Sinha J., Lua K.B., **Dash S. M.**, Physics of Fluids, 33, 081912 (2021) (Editor's Pick, Featured Article)
4. A Numerical Study on the Lid-Driven Cavity with Power-Law Fluids at Different Moving Lengths of the Top Lid by Sikdar P., **Dash S. M.**, CFD Letters, 12, 107-117 (2020)
5. A numerical study on the enhanced drag reduction and wake regime control of a square cylinder using dual splitter plates by **Dash S. M.**, Triantafyllou M. S., Alvarado P. Vy., Computers and Fluids, 199, 104421 (2020)
6. A flexible forcing immersed boundary simplified lattice Boltzmann method for two and three-dimensional fluid-solid interaction problems by **Dash S. M.**, Computers and Fluids, 184, 165-177 (2019)
7. A study on natural convection in a cold square enclosure with two vertical eccentric square heat sources using IB-LBM scheme by **Dash S. M.**, Sahoo S., Journal of Thermal Science and Engineering Applications, 11, 051013 (2019)
8. Enhanced thrust performance of a two-dimensional elliptic airfoil at high flapping frequency in a forward flight by **Dash S. M.**, Lua K. B., Lim T. T., Yeo K. S., Journal of Fluids and Structures, 76, 37-59 (2018)
9. Experimental investigation of turbulent wave boundary layers under irregular coastal waves by Yuan J., **Dash S. M.**, Coastal Engineering, 128, 22-36 (2017)
10. On the thrust performance of a flapping two-dimensional elliptic airfoil in a forward flight by Lua K. B., **Dash S. M.**, Lim T. T., Yeo K. S., Journal of Fluids and Structures, 66, 91-109 (2016)
11. Thrust enhancement on a two-dimensional elliptical airfoil in a forward flight by **Dash S. M.**, Lua K. B., Lim T. T., International Journal of Aerospace and Mechanical Engineering, 10, 265-272 (2016)
12. Natural convection in a square enclosure with a square heat source at different horizontal and diagonal eccentricities by **Dash S. M.**, Lee T. S., Numerical Heat Transfer, Part A: Applications, 68, 686-710 (2015)
13. Particle sedimentation in a constricted passage using a novel flexible forcing IB-LBM scheme by **Dash S. M.**, Lee T. S., Huang H., International Journal of Computational Methods, 12, 1350095 (2015)
14. Two spheres sedimentation dynamics in a viscous liquid column by **Dash S. M.**, Lee T. S., Computers and Fluids, 123, 218-234 (2015)
15. A flexible forcing three dimension IB-LBM scheme for flow past stationary and moving spheres by **Dash S. M.**, Lee T. S., Huang H. , Lim T. T., Computers and Fluids, 95, 159-171 (2014)
16. A novel flexible forcing hybrid IB-LBM scheme to simulate flow past circular cylinder by **Dash S. M.**, Lee T. S., Huang H., International Journal of Modern Physics C, 25, 1340014 (2014)
17. Natural convection from an eccentric square cylinder using a novel flexible forcing IB-LBM method by **Dash S. M.**, Lee T. S., Huang H., Numerical Heat Transfer, Part A: Applications, 65, 531-555 (2014)
18. Natural convection from an inclined square cylinder using novel flexible forcing IB-LBM approach by **Dash S. M.**, Lee T. S., Huang H., Engineering Applications of Computational Fluid Mechanics, 8, 91-103 (2014)

19. A novel flexible forcing hybrid IB-thermal LB model for natural convection from a circular cylinder by **Dash S. M.**, Lee T. S., Huang H., International Journal of Dynamics of Fluids, 9, 1-15 (2013)
20. Impulsively started flow topology around tandem arrangement of two square cylinder at incidence by **Dash S. M.**, Lee T. S., International Journal of Modern Physics: Conference Series, 19, 100-108 (2012)

List of Conferences Published

1. Mitigation of Thrust Deterioration at High Flapping Frequencies of a Two-dimensional Elliptic Flapping Airfoil Using Asymmetric Flapping Strokes in the Forward Flight by Sinha J., Roy S., **Dash S. M.**, 75th Annual Meeting of the APS Division of Fluid Dynamics, USA - (2022)
2. A Numerical Study on Three-Dimensional Flapping Dragonfly Wings with Optimized Input Kinematics for hovering and forward flight by Anand K., **Dash S. M.**, Armanini S., 75th Annual Meeting of the APS Division of Fluid Dynamics, USA - (2022)
3. On the aerodynamic performance of the leading-edge tubercle elliptic flapping airfoil in forward flight condition by Pinapatruni G. V., **Dash S. M.**, Sinha J., Lua K. B., 75th Annual Meeting of the APS Division of Fluid Dynamics, USA - (2022)
4. Mitigation of Thrust Deterioration at a High Flapping Frequency of a 2D Airfoil in Forwarding Flight Condition Using Asymmetric Flapping Strokes by Sinha J., **Dash S. M.**, International Mechanical Engineering Congress and Exposition, USA - (2022)
5. Effects of Flapping Frequency on the Aerodynamic Performance of the Elliptical Tandem Flapping Wings by Ranjan R., Singh A., Sinha. J., **Dash S. M.**, 9th International and 49th National Conference on Fluid Mechanics and Fluid Power, India - (2022)
6. Characterisation of the New Open Surface Recirculating Water Tunnel Facility at the Indian Institute of Technology Kharagpur by Pinapatruni G. V., Ranjan R., Charan D., Mishra S., **Dash S. M.**, 9th International and 49th National Conference on Fluid Mechanics and Fluid Power, India - (2022)
7. Investigation of the Flow Physics in an Oscillating Lid-driven Cavity with a Concentric Square Obstacle using the Lattice Boltzmann Method by Sikdar P., **Dash S. M.**, Sinhamahapatra K. P., 9th International and 49th National Conference on Fluid Mechanics and Fluid Power, India - (2022)
8. A Numerical Study on the Effects of Wing Spacing on the Thrust Performance of the Two-Dimensional Tandem Flapping Wings for Different Rear Wing Sizes by Late Nishanth S, Sinha J., Chavda S.D, **Dash S. M.**, 48th National Conference on Fluid Mechanics and Fluid Power, India - (2021)
9. Effects of the Pivot Point Locations on the Propulsive Performance of a Two-Dimensional Flapping Elliptic Airfoil with a Pitching Angular Offset by Sinha J., **Dash S. M.**, Lua K. B., 48th National Conference on Fluid Mechanics and Fluid Power, India - (2021)
10. Effects of the Moving Lid Length and Direction of Thermal Gradient on the Heat Transfer Characteristics of a Square Top-Lid Driven Cavity: A Numerical Study using Lattice Boltzmann Model by Bhunia A., **Dash S. M.**, 48th National Conference on Fluid Mechanics and Fluid Power, India - (2021)
11. A Numerical Study on the Negative Lift and Point of Non-linearity in Lift Curve of NACA 0012 Airfoil at Low Reynolds Number by Gangadhar V. R. P., **Dash S. M.**, Sinha J., Sinhamahapatra K. P., 8th International Conference on Theoretical Applied Computational and Experimental Mechanics, Kharagpur, India - (2021)
12. Aerodynamic Performance of a Two-dimensional Flapping Elliptic Airfoil in Ground Proximity by Sinha J., **Dash S. M.**, 12th International Conference on Mechanical and Aerospace Engineering, Virtual Conference - (2021) (Best Paper Presentation Award)
13. Effect of the rear wing size on the thrust performance of the two-dimensional tandem flapping wing by Nishanth S., **Dash S. M.**, Lua K. B., 11th International Conference on Mechanical and Aerospace Engineering, Greece - (2020)
14. A numerical study on the drag reduction of a circular cylinder at low Reynolds number with two contra-rotating control cylinders by Bhunia A., **Dash S. M.**, 8th International and 47th National Conference on Fluid Mechanics and Fluid Power, India - (2020)
15. Characterisation of steady flow regime and drag force on the forward and backward facing trapezoidal cylinders: A numerical study by Bhunia A., Sikdar P., **Dash S. M.**, Lua K. B., 46th National Conference on Fluid Mechanics and Fluid Power, India - (2019)
16. Lattice Boltzmann simulations of a lid-driven cavity at different moving lengths of the top lid by Sikdar P., **Dash S. M.**, Sinhamahapatra K. P., 46th National Conference on Fluid Mechanics and Fluid Power, India - (2019)
17. Effect of the pivot point locations on the wake dynamics and thrust performance of a flapping elliptic airfoil: A numerical study by Sinha J., Nishant S., **Dash S. M.**, 46th National Conference on Fluid Mechanics and Fluid Power, India - (2019)
18. Effect of the rear wing size on the thrust performance of the two-dimensional tandem flapping wing by **Dash S. M.**, Nishanth S., Sinha J., Lua K. B., 72nd Annual Meeting of the APS Division of Fluid Dynamics, USA - (2019)
19. A study on the wake regime control and drag reduction using single splitter plate for a flow past a semicircular cylinder by **Dash S. M.**, Chavda S. D., Lua K. B., International Conference on Recent Innovations and Developments in Mechanical Engineering, India - (2018) (Best Paper Award)

20. Control of wake vortex street behind a square cylinder using surface travelling waves by **Dash S. M.**, Triantafyllou M. S., Alvarado P. Vy., 70th Annual Meeting of the APS Division of Fluid Dynamics, USA - (2017)
21. Thrust enhancement on a two-dimensional elliptical airfoil in a forward flight by **Dash S. M.**, Lua K. B., Lim T. T., 18th International Conference on Aerodynamics and Fluid Mechanics, Malaysia - (2016)
22. On the thrust performance of a 2D flapping foil in a forward flight condition by **Dash S. M.**, Lua K. B., Lim T. T., 68th Annual Meeting of the APS Division of Fluid Dynamics, USA - (2015)
23. A novel flexible forcing IB-LBM scheme to simulate flow past moving sphere by **Dash S. M.**, Lee T. S., Huang H., APCOM and ISCM, Singapore - (2013)
24. Efficient hybrid IB-LBM scheme to simulate flow past circular cylinder by **Dash S. M.**, Lee T. S., Huang H., 21st International Conference on Discrete Simulation of Fluid Dynamics, India - (2012)
25. Impulsively started flow topology around tandem arrangement of two square cylinders at an incidence by **Dash S. M.**, Lee T. S., 4th International Symposium on Physics of Fluids, China - (2011)

List of Book Chapters

1. Pinapatruni G. V., **Dash S.M.**, Sinha J., Sinhamahapatra K. P., (2022) A numerical study on the negative lift and point of non-linearity in lift curve of NACA 0012 airfoil at low number. In: Ghosh A., Sinhamahapatra K. P., Joarder R., Hota S., (eds) Aerospace and Associated Technology, Taylor and Francis Group, London.
2. Bhunia A., Sikdar P., **Dash S.M.**, Lua K.B., (2021) Characterisation of Steady Flow Regime and Drag Force on the Forward and Backward Facing Trapezoidal Cylinders: A Numerical Study. In: Prabu T., Viswanathan P., Agrawal A., Banerjee J., (eds) Fluid Mechanics and Fluid Power. Lecture Notes in Mechanical Engineering. Springer, Singapore.
3. Sinha J., Sreedharan N., **Dash S.M.**, (2021) Effect of the Pivot Point Locations on the Wake Dynamics and Thrust Performance of a Flapping Elliptic Airfoil: A Numerical Study. In: Prabu T., Viswanathan P., Agrawal A., Banerjee J., (eds) Fluid Mechanics and Fluid Power. Lecture Notes in Mechanical Engineering. Springer, Singapore.
4. Sikdar P., **Dash S.M.**, Sinhamahapatra K.P., (2021) Lattice Boltzmann Simulations of a Lid-Driven Cavity at Different Moving Lengths of the Top Lid. In: Prabu T., Viswanathan P., Agrawal A., Banerjee J., (eds) Fluid Mechanics and Fluid Power. Lecture Notes in Mechanical Engineering. Springer, Singapore.
5. **Dash S.M.**, Chavda S.D., Lua K.B., (2020) A Study on the Wake Regime Control and Drag Reduction Using Single Splitter Plate for a Flow Past a Semicircular Cylinder. In: Biswal B., Sarkar B., Mahanta P., (eds) Advances in Mechanical Engineering. Lecture Notes in Mechanical Engineering. Springer, Singapore.

नॉर्दर्न कोलफील्ड्स लिमिटेड
(मिनी रत्न कंपनी)
(कोल इण्डिया लिमिटेड कीअनुषंगी कंपनी)

Northern Coalfields Limited
(A Mini Ratna Company)
(A subsidiary of Coal India Limited)

निदेशक (तकनीकी/संचालन)
CIN- U10102MP1985GOI003160
An ISO 9001, ISO 14001 & ISO 45001 Certified Company
पोस्ट-सिगंरौली कोलियरी, जिला- सिगंरौली,म. प्र., पिन 486889 / Post- Singrauli Colliery, Distt- Singrauli, M.P. PIN-486889
Phone: 07805- 266615, email: dto.ncl@coalindia.in website : www.nclcil.in

No. NCL/SGR/DTO/F-11/2024/148 Date: 16 .09.2024

ENDORSEMENT FORM

Project Title: An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: A Case Study on the Dragline

Principal Implementing Agency: Indian Institute of Technology, Kharagpur

Principle Investigator: Shri Aurobinda Routray, Professor, Electrical Engineering, IIT Kharagpur

Co-Investigators:

(a) Prof. Mamata Jenamani, Industrial Systems Engineering, IIT Kharagpur
(b) Prof. Sunita Mishra, Mining Engineering, IIT Kharagpur
(c) Prof. Sunil Manohar Dash, Aerospace Engineering, IIT Kharagpur

Objectives as per proposal:

(a) **Modelling:** Conducting a comprehensive study of the system and creating a detailed model for the system on the basis of electromagnetics, thermal, strain-field and statistical models to generate data.
(b) **Hardware Design:** Design and implementation of the sensors and the communication network and deploying a signal processing strategy to gather and analyze data from the system.
(c) **Software Design:** Developing AI and machine learning algorithms for event detection, diagnosis, prognosis and RUL prediction of faults

Benefits of the proposal to the Coal Industry:

(a) In situ real-time health monitoring
(b) Efficient condition-based maintenance scheduling Reduction of equipment down-time
(c) Foreseeing the failures and planning accordingly
(d) The real-time data can be used to corroborate production volumes and employee engagements
(e) Localization of fault and ease of diagnosis due to the availability of time-stamped data
(f) Development of accurate degradation models and estimation of RUL.

Source of Funding: Ministry of Coal S&T Grant

Further, following details are hereby provided from NCL as requested by Investigator vide email dtd 22.07.2024,

a) Certified that, NCL intends to undertake above project as sub-implementing agency, with Prof. Aurobinda Routray, Department of Electrical Engineering, IIT Kharagpur.

b) The PI will bear the responsibility of ensuring that all project activities adhere to safety guidelines, rules, regulations, laws as applicable to coal mines and any pertinent DGMS Safety Directives & Permissions.

c) NCL will not be responsible for any mishap concomitant to the project.

d) NCL will regularly be kept intimated with the progress of the project work.

e) Jayant Project of NCL has been identified as the site for the said project.

f) Excavation Department, NCL HQ along with Jayant Project will coordinate for the implementation of said project.

g) NCL will not undertake any financial and other management responsibility of the project.

h) The project's finding will not be shared with agencies outside the country or with any agency that might directly or indirectly be disadvantageous/detrimental to the business objectives of NCL in anyway.

16/09/2024

(Jitendra Malik)
Director (Technical/Operations)
NCL, Singrauli

Date: 16-09-2024

Place: Singrauli

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
KHARAGPUR - 721302, INDIA

भारतीय प्रौद्योगिकी संस्थान खड़गपुर
खड़गपुर ७२१३०२, भारत

Office of the *Dean,*
Research & Development (R & D)

ENDORSEMENT FROM THE HEAD OF THE INSTITUTION

Project Title : AI Enabled Industry-4.0 Information Architecture for Fault Diagnosis, Prognosis and Health Management of Mining Electrical Systems.

1. Certified that Indian Institute of Technology Kharagpur intends to undertake above project with Prof. Aurobindo Routray as project principal investigator and Prof. Sunita Mishra, Prof. Mamata Jenamani and Prof. Sunil Manohar Dash as project Co-Investigator.

2. Certified that necessary infrastructure facilities shall be made available to Project Team.

3. Certified that equipment proposed to be procured from the Coal SnT are not readily available in Indian Institute of Technology Kharagpur for the purpose.

4. The company/Institute assumes to undertake the financial and other management responsibilities of the project.

Debalay Chakrabarti 18/10/2024
Signature of the Head of the Institution
Prof. Debalay Chakrabarti
Associate Dean (R & D)
SRIC, IIT Kharagpur

Date : 18.10.2024
Place : IIT Kharagpur

Phone : +91-3222-282037 (O) **Fax :** +91-3222-282771 **E-mail :** deanrd@iitkgp.ac.in, deansr@hijli.iitkgp.ac.in

AUROMA ANALYTICS PRIVATE LIMITED

Visibility till the last mile

CIN: U72900WB2021PTC246348, PAN: AAVCA4653R, TIN: CALA29376E

To
The Director
Central Mine Planning & Design Institute (CMPDI)
[Gondwana Place, Kanke Road]
[Ranchi, Jharkhand - 834008]

Subject: Proposal for getting associated with the project titled "AI-Enabled Industry 4.0 Information Architecture" project to be sponsored by CMPDI to IIT, Kharagpur

Dear Sir,

As a startup, Auroma Analytics is eager to collaborate on the upcoming project titled 'AI-Enabled Industry 4.0 Information Architecture'. Should the project succeed, Auroma commits to overseeing the manufacturing and production of the devices developed through this initiative.

About Auroma Analytics:

Auroma Analytics was founded with the vision of revolutionizing industrial operations through the integration of AI and Industry 4.0 technologies. Our team comprises experts in machine learning, data analytics, and industrial automation, dedicated to developing innovative solutions to various industrial problems.

We look forward to the possibility of working together towards mutual success.

Sincerely,

Mamata Jenamani

Prof. Mamata Jenamani
Professor
Department of Industrial
& Systems Engineering
I.I.T. Kharagpur - 721302

Co-Founder: Auroma Analytics Private Limited

Dated: 2.10.2024

FORM – IX

Details of Equipment other than Computer Hardware / Software already procured under S&T Scheme of Ministry of Coal / R&D fund of CIL in the past - which are related to the below mentioned S&T project

Name of the Project: **None**

Project Code:

Principal Implementing Agency(s):

Sub Implementing Agency(s):

Sl.No	Details of Equipment	No. of Sets	Make & Model	Year of Procurement	Name of the S&T/R&D Project against which procured	Status of equipment	Remark

Note:

(i) A list of all equipment procured for the last seven years from S&T/R&D Fund giving the status and its present utilisation.
(ii) The status should cover whether the equipment is in working condition or under breakdown. If under breakdown, can it be repaired & used in this research project?
(iii) In the light of the above please give detailed justification for procurement of additional equipment when such equipment had been procured earlier under other S&T/R&D funding.
(iv) This form should be signed by Project Leader and Project Coordinator and to be incorporated in the project proposal.

FORM – X

Details of Computers, Software and accessories already procured under S&T Scheme of Ministry of Coal / R&D fund of CIL in the past - which are related to the below mentioned S&T project

Name of the Project: **None**

Project Code:

Principal Implementing Agency(s):

Sub Implementing Agency(s):

Sl.No	Details of Computers/software	No. of Sets	Make & Model	Year of Procurement	Name of the S&T/R&D Project against which procured	Present condition of the equipment/instrument	Remarks

Note:

(i) A list of all computer & accessories procured for the last three years from S&T/R&D Fund giving the status and its present utilisation.

(ii) The status should cover whether the computer & accessories are in working condition or under breakdown. If under breakdown, can it be repaired & used in this research project?

(iii) In the light of the above please give detailed justification for procurement of additional equipment when such equipment had been procured earlier under other S&T/R&D funding.

(iv) This form should be signed by Project Leader and Project Coordinator and to be incorporated in the project proposal.

FORM – XI

DETAILS OF MANPOWER COST (SALARY AND WAGES)

Name of the Project: **An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: A Case Study on the Drag Line**

Project Code: **New**

Principal Implementing Agency(s): **Indian Institute of Technology Kharagpur**

Sub Implementing Agency(s): **Northern Coalfield Limited, India**

Sl. No.	Designation	No. of Persons	Total No. of months required/persons	Salary/month, InRs.	Total salary (3 x 4 x 5), In Rs.	Year-wise phasing, InRs.		
1	2	3	4	5	6	1st Yr.	2nd Yr.	3rd Yr.
1.	Senior Research Fellow	4	24	45,000/-	43,20,000	21,60,000	21,60,000	-
2.	Field Worker	2	24	20,000/-	9,60,000	4,80,000	4,80,000	-

Note:

- Engagement of the personnel (Research Associate/Research Assistant/Research Fellow/JRF/SRF) for implementation of the project shall be the responsibility of the Principal Implementing or Sub-implementing Agency(s) as the case may be. The payment to be made to the engaged personnel for the desired duration of the project shall be within the relevant Norms/Guidelines of the concerned related Principal Implementing or Sub-implementing Agency(s).
- Salaries of permanent employees of the Principal Implementing / Sub-implementing Agency(s) are not admissible
- The form should be jointly signed by Project Leader/Coordinator and Associated Finance Officer of the Principal Implementing /Sub-implementing Agency(s) and to be incorporated in the project proposal.

FORM – XII

DETAILS OF TRAVEL EXPENDITURE (TA/DA)

Name of the Project: **An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: A Case Study on the Drag Line**

Project Code: **New**

Principal Implementing Agency(s): **Indian Institute of Technology Kharagpur**

Sub Implementing Agency(s): **Northern Coalfield Limited, India**

Designation	From (Place To Place)	Approx. Distance, In Km	Mode	Fare, In Rs.	No. of Trips	Total travel Expenses (5 X 6), In Rs.	No. of days Involved for Daily Allowance (DA), In Rs.	Rate of DA per day, In Rs.	Total DA (6 x 8 x 9), In Rs.	Total TA+DA (7+10), InRs.
1	2	3	4	5	6	7	8	9	10	11
Professor/ Associate Professor/ Assistant Professor	Kharagpur to Singrauli, Madhya Pradesh	1010	By Air and by Road, Accommodation	Rs. 40,000/-	8	Rs. 4,00,000/-	8-10	Rs. 1,500	Rs. 1,50,000/- (Approx.)	Rs. 5,50,000/-
Project Staff (SRF/Field Worker)	Kharagpur to Singrauli, Madhya Pradesh	625	By train	Rs. 20,000/-	10	Rs. 2,00,000/-	All TA and DA included	-	-	Rs. 2,00,000/-

NB:

- The form should be jointly signed by Project Leader/Coordinator and Associated Finance Officer of the Principal Implementing Agency(s)/Sub-implementing Agency(s) and to be incorporated in the project proposal.